PIO36 User Manual

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Revision	**Comments**
0	Original Issue
1	Updated register table
2	
3	Updated register table. Added information about connector face views in 5.10. Corrected alternative Harwin connector in 5.10.3. Updated 3, 5.1, 5.10.2, 5.8, 6 and 7. Added Ethernet Phy config and RSTP enable.
4	Added description of diagnostics signal in 4.3. Updated register table, clarified alarm usage. Renamed Ehternet Phy to Ethernet Port.

1 General

1.1 Scope of document

This document provides detailed information required for system integration of the Innova PIO36 board. The purpose of the document is to provide a clear understanding of the features of the device as well as the operation limits and interface requirements.

1.2 References and abbreviations

1.2.1 References

Ref.	Document ID	Rev.	Date	Document Name
1	NA	V1.1b3	26.04.2012	MODBUS APPLICATION PROTOCOL SPECIFICATION V1.1b3
2	NA	V1.0b	24.10.2006	MODBUS Messaging on TCP/IP Implementation Guide V1.0b
3	NA	NA	29.08.2008	IEEE 754-2008 – IEEE Standard for Floating-Point Arithmetic
4	NA	NA	06.07.2018	IEEE 802.1Q-2018 - IEEE Standard for Local and Metropolitan Area Networks–Bridges and Bridged Networks

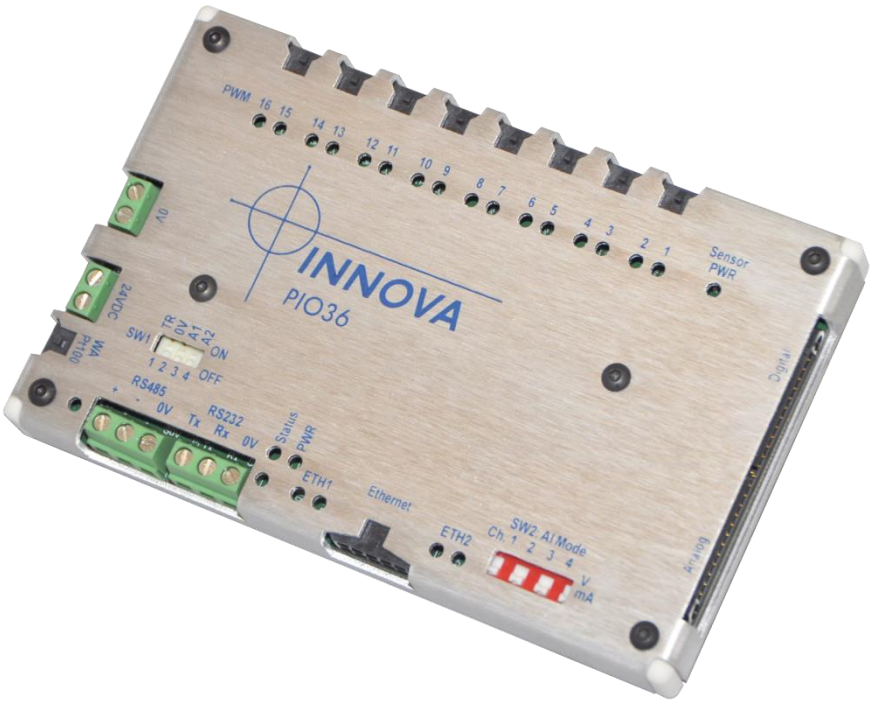
1.2.2 Abbreviations

Mbps	Mega bit per second
EMC	Electro Magnetic Compatibility
ESD	Electrostatic Discharge
I/O	Input / Output
inf	Infinity
NaN	Not a Number
PCB	Printed Circuit Board
PWM	Pulse Width Modulation
RIP	Remote Intervention Protocol
RSTP	Rapid Spanning Tree Protocol
SW	Software
WA	Water ingress Alarm

1.3 ESD precautions

The unit contains components that are sensitive to the high voltages that can be generated by the human body due to static charges. To prevent ESD damage, the unit must be stored and handled in accordance with IEC 61340-5-1: Protection of electronic devices from electrostatic phenomena – General requirements.

2 Description



The PIO36 board is designed to control 16 individual proportional 24 VDC loads. The design allows the unit to be installed in a pressurised environment e.g., compensated submerged oil-filled enclosure. Typical applications are solenoid or proportional hydraulic valve operation for marine/subsea environments. Proportional control is possible by pulse width modulated (PWM) control of the load. The unit has the following I/O:

- 16x PWM outputs for driving solenoids.
- 16x analogue inputs 4 to 20 mA signal, for instance pressure or temperature transmitters.
- 4x isolated digital inputs are available and can be used to monitor end switches or other on/off conditions, or to count pulses or gray2 encoder signals.
- 16x power outputs for loop powered 4 to 20 mA transmitters.
- 4x power outputs for analogue / digital sensors.
- 1x input for Pt100 temperature sensor input.
- 1x connection for water alarm probe.

Remote control is possible via Modbus TCP over Ethernet or Modbus RTU over either RS232 or multi drop RS485 communication.

The unit is enclosed in an aluminium cover with marking for the terminal connections.

3 Features

Power

Power supply: 10 – 30 VDC, 10 A max.

Output

16 x 24 V PWM or on/off function 2 A each (maximum 10 A simultaneous load for all channels), 16-bit resolution 0 to 100%

LED indicators on all PWM outputs

Input

16 x Analogue inputs 4 to 20 mA 16-bit resolution, 4 of these channels can be configured for 0 – 10 V.

16 x current limited (max 25 mA) power outputs for 4 to 20 mA transmitters.

4 x isolated digital inputs 12 – 24 V, with pulse counting.

4 x power outputs for digital / analogue sensors.

Water ingress alarm

Temperature reading from Pt100 sensor.

Communication

Ethernet, RS232 or RS485 communication

Fail safe in case of communication error (user defined state)

Environment

Designed to work submerged in oil at a pressure of up to 400 bar.

Qualified according to API17F chapter 9.3.2 for temperature, vibration, and shock.

4 Functional description

4.1 Concept

The PIO36 controller is a slave controller designed to execute commands as requested by a master system. All physical I/O is available from the master via Modbus TCP or RTU. The board is designed as a single compact unit.

4.2 Communication interfaces

The PIO36 can communicate with a master by Ethernet, RS232 or RS485 interface.

4.2.1 Ethernet interface

The Ethernet interface is usable up to 100 m distance. The PIO36 supports 10 Mbps and 100 Mbps, it has 2 ports with an internal switch that allows easy daisy chaining of several units. The internal switch has support for Rapid Spanning Tree Protocol (RSTP).

4.2.2 RS232 interface

The RS232 interface can only be used in the event where the master or node is within the range of the RS232 limits (15 m). The physical connection of RS232 does not allow multiple controllers on the same port. A local master may however use multiple RS232 ports to control several PIO36 slave units.

4.2.3 RS485 interface

The RS485 interface allows extended distance between the master and slave (up to 1000 m) as well as multi drop capability (several units on the same line). Up to 32 units can be connected to a RS485 line. However, having multiple units on the same line will slow down the responses of the units connected.

4.2.4 Software and communication

Communication between the master and the slave (PIO36) are either Modbus TCP over Ethernet or Modbus RTU over either RS232 or multi drop RS485 communication.

4.3 PWM output channels

The PWM output channels are based on high side drivers feeding the load with 24 V power on fixed frequency square wave. The load is connected between the PIO36 output and 0 V. The duty cycle can be individually varied from the master controller. The frequency can be set from 1 to 500 Hz. The PWM frequency is set in groups of 4 channels, so it is possible to utilize different PWM frequency for different types of hydraulic valves. This feature allows corrections in the event of valve oscillation etc.

The duty cycle has a resolution of 16-bits and is set as a value between 0 to 100%.

The output channels are configured with freewheeling diode protection to reduce the voltage over an inductive load when switching. However, any inductive loads connected should have local freewheeling diode mounted near the coil. This will reduce the noise level as well as heat generated on the board. Innova SVB boards (PN 1041586) contain freewheeling diode and LED indicating valve status and are designed for this purpose.

The board utilize an "intelligent" transistor for the PWM and sensor power outputs. This includes short circuit protection, thermal protection and a diagnostic feedback signal. The diagnostic feedback has different meaning depending on the state of transistor (ON or OFF). When the transistor is OFF the diagnostic feedback indicate detection of open load (or too low load). In many cases this is caused by a wire or solenoid break. When the transistor is ON the diagnostic feedback indicates thermal shutdown of the transistor. The diagnostic feedback is used to trigger the Fault Flags described in chapter 10.

4.4 Isolated digital inputs

Four optically isolated digital inputs are available. These inputs accept a 12 – 24 VDC input level.

4.4.1 Counters

The digital inputs count number of rising edges on digital input 1-4. Additionality digital input 1 and 2 counts as a gray2 encoder under encoder 1, and digital input 3 and 4 counts as a gray2 encoder under encoder 2.

4.4.2 Counter frequency

Each counter also calculates the frequency of the input signal by checking the number of pulses over the last second.

4.5 Analogue input channels

The PIO36 has 16 analogue inputs with 16-bit resolution. These inputs read 0 to 20 mA and are designed to accept 4 to 20 mA signals typically from pressure/temperature transmitters. 4 of these input channels can be configured to 0 to 10 V voltage inputs instead of 0 to 20 mA by switches (SW2).

5 Connections

The figure below shows the layout of the PIO36 terminals.

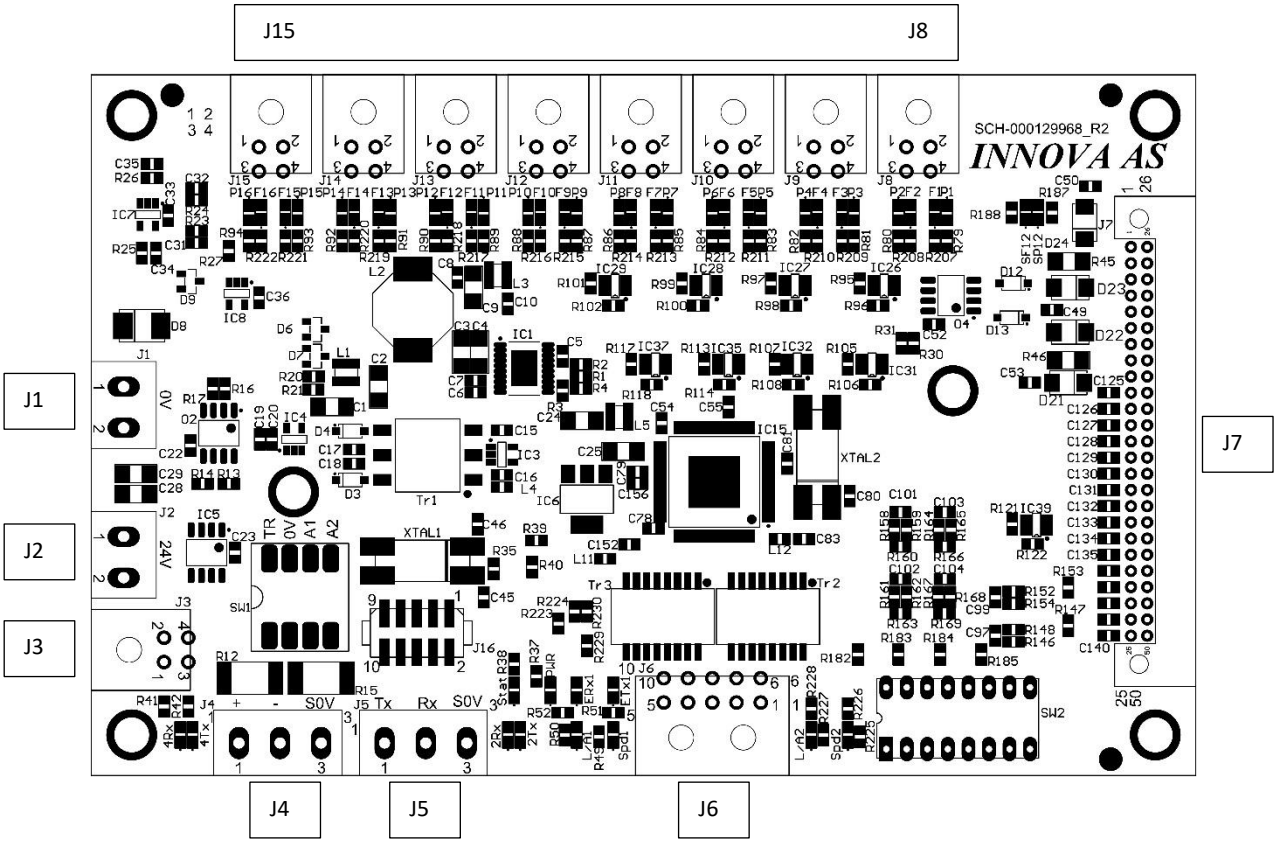


Figure 1

The following terminals are available on the PIO36:

Power	Screw terminal (J1 & J2)
RS232	Screw terminal (J5)
RS485	Screw terminal (J4)
Ethernet	Connector (J6)
PWM outputs (1 - 16)	Connectors (J8 - J15)
Analogue and Digital Inputs	Connector (J7)
WA and Pt100	Connector (J3)

Cables for the signal inputs and PWM outputs should be kept as short as possible and certainly less than 3 m.

5.1 Power

Power consumption depends on type of load, number of PWM channels driven and maximum duty cycle. Make sure to use a 24 VDC power supply capable of supplying the necessary power. Use sufficiently thick wires for 24 VDC and 0 V connections when using high power output from the PIO36. Power supply 0 V connects to terminal J1 which has 2 terminals. Power supply 24 V connects to J2, also

with 2 terminals. Use both terminals for the power connections when using high power loads on the PWM outputs.

The PIO36 is designed to be supplied form an AC to DC power supply. If it is supplied from a DC distribution network additional protection is required to comply with EMC regulations.

5.2 RS232

The RS232 port is configured with the screw terminals on connector J5.

The default communication parameters are Baud rate 115200, 8 data bits, No parity and 1 stop bit. No handshake signals are used.

5.3 RS485

The RS485 is connected to screw terminal J4. The RS485 line should carry a separate ground wire (G) terminated at the RS485 port in both ends. The + and – signal should be carried in a twisted pair with a characteristic impedance close to 120 Ω. This is especially important when communicating over a long distance.

The RS485 port on the PIO36 is galvanic isolated. The shield should be terminated at the master protective ground. This configuration has proved to be sufficient to prevent problems due to different ground levels at master and PIO36 and provide protection against large transients.

A RS485 line must be terminated with a termination resistance (120 Ω) at each end. This can be done on the PIO36 by setting dip switch marked TR on the PIO36 cover (SW1A) to the on position (as shown on the switch). On a line with several nodes a terminal resistor is required in each end of the line. Normally this is on the master (usually at one end) and at the node at the most remote end of the line, all the other nodes should not have termination resistors fitted. If the line has only a master and one slave, both should have a termination resistance.

If there is no individual 0 V for RS485 the switch marked 0V on the PIO36 cover (SW1B) must be set to the ON position (as shown on the switch) to connect local 0 V to RS485 0 V.

The default communication parameters are Baud rate 115200, 8 data bits, no parity and 1 stop bit.

5.4 Ethernet

2 Ethernet ports are provided at connector J6. Use suitable ethernet cables for 10/100 Mbit and keep the twisting of the pairs as close as possible to the connector.

5.5 PWM outputs

The 16 PWM outputs are numbered 1, 2, ..., 16. The PIO36 uses high side drivers, supplying 24 V during the on-time of the PWM pulse. If the duty cycle is set to 100% the output is continuously ON and can be used as a power output for sensors, lights or similar loads not designed for PWM. The external load must be connected between the PIO36 output and 0 V. The PIO36 connectors for PWM output are 4 pins, with 2 pins for 2 output channels and 2 pins for 0 V.

5.6 Analogue inputs

The analogue inputs are designed for accepting 4 to 20 mA signals, typically from pressure/temperature transmitters. The analogue inputs are connected to connector J7 (also used for digital inputs).

Inputs 1 to 4 can be configured for 0 to 10 V inputs signals instead of 4 to 20 mA by switching the corresponding switch (SW2A to SW4B) to the on position (as shown on the switch).

5.7 Digital inputs

The digital inputs are galvanic isolated from the PIO36 by use of optocouplers. The input contains current limiting resistors which makes the inputs suitable for 12 to 24 VDC input level. The digital inputs are connected to connector J7 (also used for analogue inputs).

5.8 Status LEDs

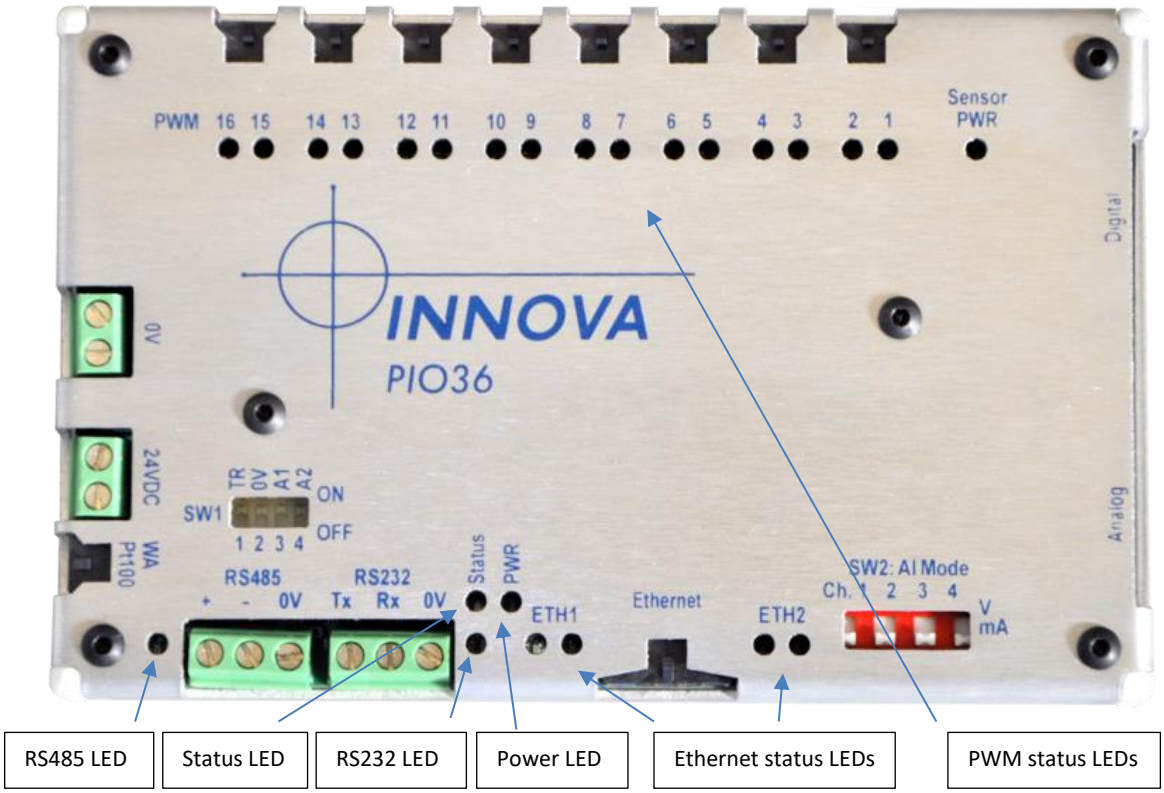
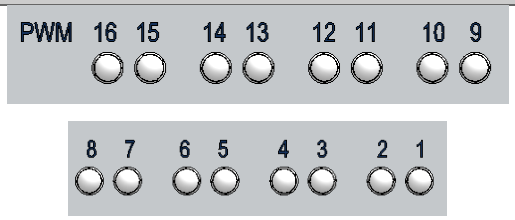


Figure 2 - Valve driver board LED indicators

The PIO36 valve driver board uses the following LEDs:

Marking		Description
PWR		• Continuous green – 24 VDC supply power OK
Status		• Blinking 1 Hz red – SW running OK
Sensor PWR 1)		• Continuous green – 24 VDC power output to J7 is On • Continuous red – 24 VDC power output to J7 detects open circuit (when power is off) or the load is short circuited

Marking		Description
ETH1		Left LED: • Continuous green, link • Blinking green, activity on Ethernet Right LED: • Off, 10 Mbit/s • Continuous green, 100 Mbit/s
ETH2		Left LED: • Continuous green, link • Blinking green, activity on Ethernet Right LED: • Off, 10 Mbit/s • Continuous green, 100 Mbit/s
RS232		• Green LED: Tx • Yellow LED: Rx
RS485		• Green LED: Tx • Yellow LED: Rx
PWM 1 to 16 [1)]		• No light – Output deactivated, and a load is present on output • Continuous green – Output activated • Continuous red – Fault, open circuit (when the output is inactive) • Continuous orange (green and red) – Fault, short circuit

1) The sensor power and PWMx LED consists of two LEDs, one green and one red.

5.9 DIP switches

The PIO36 has 2 DIP switches with 4 switches each, these are summarised in the tables below:

SW1 DIP switch	
SW1A	Enable / disable RS485 termination
SW1B	Connects isolated RS485 0 V to PIO36 input power 0 V
SW1C	Least significant address bit
SW1D	Most significant address bit

SW2 DIP switch	
SW2A	Analogue input 1, mA or V
SW2B	Analogue input 2, mA or V
SW3C	Analogue input 3, mA or V
SW2D	Analogue input 4, mA or V

5.10 Connector pinouts

Note: All connector views in this chapter are face view of the cable connector (mating with the corresponding PIO36 connector).

5.10.1 J3 - WA and Pt100

Mating connector: Molex 43025-0400, pins: 43030-0007

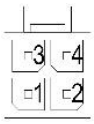


Figure 3 Face view of mating connector.

J3 - WA and Pt100	
Pin #	**Function**
1	WA probe
2	WA probe
3	Pt100
4	Pt100

The 2 WA probe pins are connected to a water detection probe designed and placed such that any sea water leaked into the enclosure will conduct an electrical current between the pins.

The Pt100 pins connects to a Pt100 resistor.

5.10.2 J6 - Ethernet

Mating connector: Molex 43025-1000, pins: 43030-0007

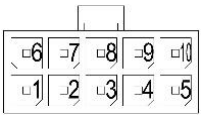


Figure 4 Face view of mating connector.

J6 - Ethernet	
Pin #	**Function**
1	Ethernet 2, Tx+
2	Ethernet 2, Rx+
3	Factory defaults
4	Ethernet 1, Tx+
5	Ethernet 1, Rx+
6	Ethernet 2, Tx-
7	Ethernet 2, Rx-
8	Factory defaults
9	Ethernet 1, Tx-
10	Ethernet 1, Rx-

If the 2 Factory defaults pins are shorted, the persistent storage is not loaded at start-up, and all values will use their default values. The user then must set all variables to their desired value, then flush the persistent storage. Or power cycle the board without the pins shorted.

5.10.3 J7 - Analogue and digital inputs

Mating connector: Nicomatic 222S50M16 or Harwin M80-48050xx or M80-48050xx (xx is pin finish, 05 or 42)

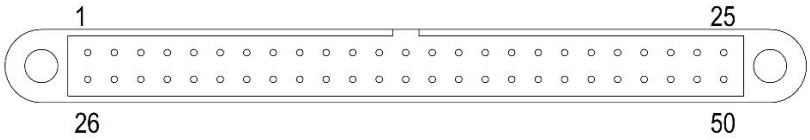


Figure 5 Face view of mating connector.

J7 - Analogue and digital inputs			
Pin #	**Function**	**Pin #**	**Function**
1	Digital input 1+	**26**	Sensor power
2	Digital input 1-	**27**	0 V
3	Digital input 2-	**28**	Sensor power
4	Digital input 2+	**29**	0 V
5	Digital input 3+	**30**	Sensor power
6	Digital input 3-	**31**	0 V
7	Digital input 4-	**32**	Sensor power
8	Digital input 4+	**33**	0 V
9	0 V	**34**	Sensor power
10	Analogue input 1	**35**	Loop power 1
11	Analogue input 2	**36**	Loop power 2
12	Analogue input 3	**37**	Loop power 3
13	Analogue input 4	**38**	Loop power 4
14	Analogue input 5	**39**	Loop power 5
15	Analogue input 6	**40**	Loop power 6
16	Analogue input 7	**41**	Loop power 7
17	Analogue input 8	**42**	Loop power 8
18	Analogue input 9	**43**	Loop power 9
19	Analogue input 10	**44**	Loop power 10
20	Analogue input 11	**45**	Loop power 11
21	Analogue input 12	**46**	Loop power 12
22	Analogue input 13	**47**	Loop power 13
23	Analogue input 14	**48**	Loop power 14
24	Analogue input 15	**49**	Loop power 15
25	Analogue input 16	**50**	Loop power 16

The digital inputs consist of opto-coupler LED with series resistors for 12 - 24 VDC, the signal is connected to the digital inputs with positive power to the + input and negative (0 V) to the - input. Often devices with open collector are used to generate signals for the digital inputs, these come with 2 types of outputs: PNP and NPN. If an PNP device is used the output is typically connected to the digital + input and the digital - input is connected to 0 V. If a NPN device is used the output is typically connected to the digital - input and the digital + input is connected to 24 V.

Analogue 4 to 20 mA outputs (or 0 to 10 V for channel 1 to 4) are connected to the analogue inputs on J7. If loop powered transmitters are used these can be supplied from the Loop power outputs on J7. The loop power output is connected to the supply input of the transmitter and the signal output from the transmitter is connected to the analogue input on J7. The supply current on the loop power outputs is limited to approximately 25 mA.

The sensor power output and 0V pins on J7 can be used to power transmitters which is not loop powered. Maximum current load on these outputs is 0.5 A per pin and 2 A total for all pins.

5.10.4 J8 - J15 - PWM outputs

Mating connector: Molex 43025-0400, pins: 43030-0007



Figure 6 Face view of mating connector.

J8 - J15 - PWM 1 - 16 outputs	
Pin #	**Function**
1	0 V
2	0 V
3	PWM output 2
4	PWM output 1

The table above shows connections for channel 1 & 2 on J8. Pinout for channels 3 & 4 are on J9 with corresponding pinout. And so on for channels 5 - 16.

The load (solenoid coil) is connected between the output and 0 V i.e., between pin 1 and 3 and between pin 2 and 4.

6 Physical layout, mounting

Size and mounting interface.

Size: 90 x 140 x 13 mm (L x W x H)

Mounting: 6 M3 bolts from the underside. Always use at least the 4 corner bolts when mounting the PIO36. See drawing below for mounting hole positions. If used in a vibrating environment use all 6 mounting points.

Housing: Aluminium cover

Always mount the PIO36 on a metal surface connected to protective earth and place it inside a suitable enclosure (IP65 or higher depending on application). Never operate the system without the PIO36 in an enclosure as accidental shorts to the screw terminals can cause unpredictable operation of the PWM outputs or damage to the PIO36 or connected equipment.

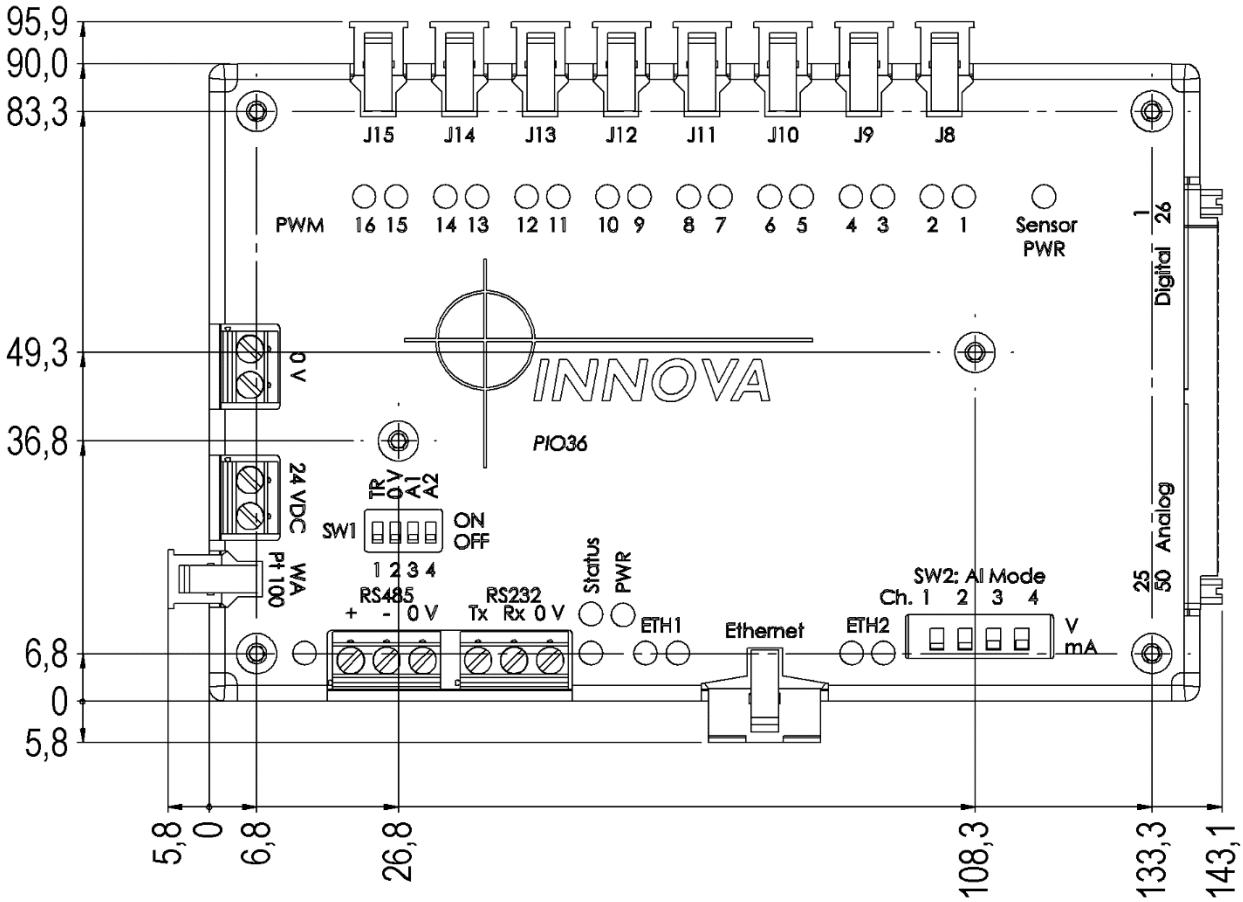


Figure 7 - Dimensions and mounting holes

7 Specification

Parameter	Value
Power supply voltage:	Nominal 24 VDC (10 – 30 VDC)
Power consumption (idle):	60 mA
Power supply current:	10 A, including all PWM and power outputs
PWM output current:	2 A each channel, 10 A total for all channels (including sensor power outputs)
PWM output resolution:	16-bit
Loop power supply:	25 mA
Sensor power supply:	0.5 A per output pin on J7, 2A total for all pins
Typical response time:	1.75 ms required Modbus RTU inter frame delay
Analogue inputs resolution:	16-bit
Analogue inputs sample rate:	500 Hz
Pt100 temperature measuring range:	-10 to 120°C
Digital input nominal input voltage:	12 to 24VDC
Max digital input frequency:	4 kHz
Digital input threshold volage:	5V
Humidity:	10 to 95% RH Non-condensing
Ambient pressure:	400 bar
Operating temperature:	-20 to 50 °C See Note 1 below
Storage temperature:	-40 to 85 °C
Vibration:	5 Hz to 25 Hz with ±2 mm displacement, 25 Hz to 1000 Hz with 5 g acceleration
Schock:	30 g acceleration, 11 ms half sine
Industry standard qualification:	according to API17F, chapter 9.3.2 for temperature, vibration, and shock

Note 1:

The actual maximum operating temperature depends on the load connected to the unit and mounting conditions. A high load, that is the combination of load on each channel and number of channels in use, give an increase in power dissipation in the output drivers. The temperature rise on the output drivers depends on the power dissipation and cooling, i.e. how the unit is mounted: in still air, circulating air or oil. Oil providing the best cooling and still air the worst. The power loss in the output driver is greatly increased when using an inductive load (such as a solenoid valve) without a dedicated freewheeling diode across the load. In this case the back EMF from the load is dissipated in output driver. It is therefore always recommended to use dedicated freewheeling diodes across the load.

The output drivers contain internal temperature shutdown to protect the drivers if the driver temperature becomes too high.

8 Configuration instruction

8.1 Ethernet

The unit is addressed directly by its IP-address and does not use the *Unit Identifier* field in the MBAP header. The *Unit Identifier* value from the request is copied to the response. See *MODBUS Messaging on TCP/IP Implementation Guide [1.2.1, 2]* for more details.

The board is supplied with the following default settings:

- IP address: 192.168.52.209
- Netmask: 255.255.248.0
- Default gateway: 192.168.48.1
- RSTP: Enabled

The IP address settings are stored in the board's Modbus register. To change the settings, use a Modbus client to update the appropriate registers. To store the new values in EEPROM, the EEPROM flush command (register 400113) must be sent. The board will then remember the new settings after a reboot. See section 10 for details.

The Innova Discover Utility can be used to discover the PIO36 on the network and change the IP configuration.

8.2 RS232 / RS485 address

The PIO36 can be set up with 4 different addresses by dip switch selector. The switches are marked A1 and A2 on the PIO36 cover. When using several PIO36 (or other devices) on a RS485 line, all units must have different addresses. The master must always address the PIO36 with the correct address, even when communicating with a single unit on RS232.

Address offset	AD1 (SW1C)	AD2 (SW1D)	
0	Off	Off	Base address (default 10)
1	On	Off	Base address + 1
2	Off	On	Base address + 2
3	On	On	Base address + 3

The base address is by default set to 10, the possible addresses set by the dip switches are then 10, 11, 12 & 13.

The base address and serial communication settings are stored in the board's Modbus register. To change the settings, use a Modbus client to update the appropriate registers. To store the new values in EEPROM, the EEPROM flush command (register 400113) must be sent. The board will then remember the new settings after a reboot. See section 10 for details.

Address detection is only performed at boot up of the system. The PIO36 need to be powered down and up again to detect changes in the status of the dip switches.

8.3 Serial communication mode

8.3.1 RS485 settings

RS485 termination resistor is switched in with the dip switch marked TR (SW1) on the PIO36 cover. Set this switch to the on position (as shown on the switch) on the unit at the end of the RS485 line, as described above.

The dip switch marked 0 V on the PIO36 cover (SW1) connects the RS485 0 V wire to 0 V in the PIO36. This is only recommended used when an individual RS485 0 V is not available.

9 Modbus

MODBUS is an application-layer messaging protocol, positioned at level 7 of the OSI model. It provides client/server communication between devices connected on different types of buses or networks.

The protocol specification and other technical documents describing the Modbus protocol are available for download at https://modbus.org/.

9.1 Supported Function Codes

The supported function codes are listed in Table 1.

Function code	Description
0x03	Read Holding Registers
0x04	Read Input Registers
0x06	Write Single Register
0x10	Write Multiple Registers

Table 1 - Supported Function Codes

9.2 Register Addressing

Register 300001 to register 365536 refer to a Modbus input register in the range 1 – 65536. Register 400001 to register 465536 refer to a Modbus holding register in the range 1 – 65536.

9.3 Datatypes

9.3.1 Uint16

Uint16 is an unsigned 16-bit word as defined in Modbus Application Protocol Specification [1].

9.3.2 Int16

Int16 is a two-complements signed 16-bit word stored in a single register.

9.3.3 Uint8

Uint8 is an unsigned 8-bit word stored in a single register. The 8 most significant bits are ignored on writes, and they read as 0.

9.3.4 Uint32

Uint32 is an unsigned 32-bit word, consisting of two sequential registers, with the MSB stored in the lowest numbered register (big-endian).

9.3.5 Int32

Int32 is a two-complements signed 32-bit word, consisting of two sequential registers, with the MSB stored in the lowest numbered register (big-endian).

9.3.6 Float

Float is a 32-bit single precision floating point type (binary32 in IEEE 754 [3]) stored in two sequential registers with the MSB stored in the lowest numbered register (big-endian).

9.3.7 IPv4 address

IPv4 addresses are stored as a 32-bit word, consisting of two sequential registers. The octets of the ip-address are stored little-endian, e.g., the IP-address 192.168.1.100 is stored the same as the Uint32 value 0x6401A8C0, the least significant byte is 192 (0xC0) and the most significant byte is 100 (0x64).

9.3.8 Addressing multi-registers values

All multi-register values are addressed using their lowest numbered register.

Note: Trying to access only part of a multi-register datatype (e.g., only one register of a float value) will result in a Modbus Illegal Data Address exception.

9.4 Gateway functionality

If the Modbus Gateway is enabled (see register 300149/400149), packets received on the RS232 interface, but not addressed to this board are transmitted on the RS485 port to allow the board to function as a gateway for other boards.

This is also true for Modbus TCP requests to port 503, those packets will be sent out on the RS485 port. If no response is detected a Modbus Gateway Target Device Failed to Respond exception (code 11) is sent as a response on the TCP connection.

Since the Modbus TCP requests are simply forwarded to the serial port the field *Unit Identifier* needs to be set correctly according to the address of the serial device.

10 Registers

Note: in the table below, a register longer than 16 bit (e.g. Uint32 values) are treated as one register, but consists of two subsequent Modbus addresses.

To write persistent changes to EEPROM (e.g. to change IP address or other communications parameters), the EEPROM flush command (register 400113) must be sent. The board will then remember the new settings after a reboot. See section 10.6.3 for details.

10.1 Table of registers

Standard Register Range			
Name	**Register**	**Datatype**	**Note**
Product Number	300001	Uint32	
Serial Number	300003	Uint32	
Hardware Version	300005	Uint32	
FW version Major	300007	Uint16	
FW version Minor	300008	Uint16	
FW version Patch	300009	Uint16	
Firmware CRC	300010	Uint32	
Configuration CRC	300012	Uint32	
Heartbeat	300014	Uint16	
Ipv4 Address	300050	IPv4 Address	
Subnet Mask	300052	IPv4 Address	
Default Gateway	300054	IPv4 Address	
MAC address[0]	300056	Uint16	
MAC address[1]	300057	Uint16	
MAC address[2]	300058	Uint16	
Baudrate RS232	300059	Uint32	
Baudrate RS485	300061	Uint32	
Modbus Base Address	300063	Uint8	
Modbus Address	300064	Uint8	
Lost-coms timeout	300065	Uint16	
Lost-coms Keep Value	300066	Uint16	
Power-on default output 1	300067	Float	
Power-on default output 2	300069	Float	
Power-on default output 3	300071	Float	
Power-on default output 4	300073	Float	
Power-on default output 5	300075	Float	

Standard Register Range			
Name	**Register**	**Datatype**	**Note**
Power-on default output 6	300077	Float	
Power-on default output 7	300079	Float	
Power-on default output 8	300081	Float	
Power-on default output 9	300083	Float	
Power-on default output 10	300085	Float	
Power-on default output 11	300087	Float	
Power-on default output 12	300089	Float	
Power-on default output 13	300091	Float	
Power-on default output 14	300093	Float	
Power-on default output 15	300095	Float	
Power-on default output 16	300097	Float	
Sensor Power configuration	300099	Uint16	
Uptime	300100	Float	
System Warnings	300102	Uint16	
Sensor Alarms	300103	Uint16	
PWM Faults	300104	Uint16	
Modbus Packet Count	300105	Uint32	
Modbus Dropped Packets Count	300107	Uint32	
EEPROM write counter	300109	Uint32	
Sensor Power Enable	300111	Uint8	
rs232_terminal	300115	Uint8	
rs485_terminal	300116	Uint8	
Lost COM's output 1	300117	Float	From version 1.3.0
Lost COM's output 2	300119	Float	From version 1.3.0
Lost COM's output 3	300121	Float	From version 1.3.0
Lost COM's output 4	300123	Float	From version 1.3.0
Lost COM's output 5	300125	Float	From version 1.3.0
Lost COM's output 6	300127	Float	From version 1.3.0
Lost COM's output 7	300129	Float	From version 1.3.0
Lost COM's output 8	300131	Float	From version 1.3.0
Lost COM's output 9	300133	Float	From version 1.3.0
Lost COM's output 10	300135	Float	From version 1.3.0

Standard Register Range			
Name	**Register**	**Datatype**	**Note**
Lost COM's output 11	300137	Float	From version 1.3.0
Lost COM's output 12	300139	Float	From version 1.3.0
Lost COM's output 13	300141	Float	From version 1.3.0
Lost COM's output 14	300143	Float	From version 1.3.0
Lost COM's output 15	300145	Float	From version 1.3.0
Lost COM's output 16	300147	Float	From version 1.3.0
Modbus Gateway enabled	300149	Bool	From version 1.3.2
PWM Channel 1 Duty	301000	Float	
PWM Channel 2 Duty	301002	Float	
PWM Channel 3 Duty	301004	Float	
PWM Channel 4 Duty	301006	Float	
PWM Channel 5 Duty	301008	Float	
PWM Channel 6 Duty	301010	Float	
PWM Channel 7 Duty	301012	Float	
PWM Channel 8 Duty	301014	Float	
PWM Channel 9 Duty	301016	Float	
PWM Channel 10 Duty	301018	Float	
PWM Channel 11 Duty	301020	Float	
PWM Channel 12 Duty	301022	Float	
PWM Channel 13 Duty	301024	Float	
PWM Channel 14 Duty	301026	Float	
PWM Channel 15 Duty	301028	Float	
PWM Channel 16 Duty	301030	Float	
PWM Ignore faults	301033	Uint16	
PWM Frequency Channels 1-4	301034	Float	
PWM Frequency Channels 5-8	301036	Float	
PWM Frequency Channels 9-12	301038	Float	
PWM Frequency Channels 13-16	301040	Float	
PWM Channel 1 Dither Duty	301042	Float	
PWM Channel 2 Dither Duty	301044	Float	
PWM Channel 3 Dither Duty	301046	Float	
PWM Channel 4 Dither Duty	301048	Float	

Standard Register Range			
Name	**Register**	**Datatype**	**Note**
PWM Channel 5 Dither Duty	301050	Float	
PWM Channel 6 Dither Duty	301052	Float	
PWM Channel 7 Dither Duty	301054	Float	
PWM Channel 8 Dither Duty	301056	Float	
PWM Channel 9 Dither Duty	301058	Float	
PWM Channel 10 Dither Duty	301060	Float	
PWM Channel 11 Dither Duty	301062	Float	
PWM Channel 12 Dither Duty	301064	Float	
PWM Channel 13 Dither Duty	301066	Float	
PWM Channel 14 Dither Duty	301068	Float	
PWM Channel 15 Dither Duty	301070	Float	
PWM Channel 16 Dither Duty	301072	Float	
PWM Channel 1 Dither Frequency	301074	Float	
PWM Channel 2 Dither Frequency	301076	Float	
PWM Channel 3 Dither Frequency	301078	Float	
PWM Channel 4 Dither Frequency	301080	Float	
PWM Channel 5 Dither Frequency	301082	Float	
PWM Channel 6 Dither Frequency	301084	Float	
PWM Channel 7 Dither Frequency	301086	Float	
PWM Channel 8 Dither Frequency	301088	Float	
PWM Channel 9 Dither Frequency	301090	Float	
PWM Channel 10 Dither Frequency	301092	Float	
PWM Channel 11 Dither Frequency	301094	Float	
PWM Channel 12 Dither Frequency	301096	Float	
PWM Channel 13 Dither Frequency	301098	Float	
PWM Channel 14 Dither Frequency	301100	Float	
PWM Channel 15 Dither Frequency	301102	Float	
PWM Channel 16 Dither Frequency	301104	Float	
PWM Channel 1 Dither Phase	301106	Float	
PWM Channel 2 Dither Phase	301108	Float	
PWM Channel 3 Dither Phase	301110	Float	
PWM Channel 4 Dither Phase	301112	Float	

Standard Register Range			
Name	**Register**	**Datatype**	**Note**
PWM Channel 5 Dither Phase	301114	Float	
PWM Channel 6 Dither Phase	301116	Float	
PWM Channel 7 Dither Phase	301118	Float	
PWM Channel 8 Dither Phase	301120	Float	
PWM Channel 9 Dither Phase	301122	Float	
PWM Channel 10 Dither Phase	301124	Float	
PWM Channel 11 Dither Phase	301126	Float	
PWM Channel 12 Dither Phase	301128	Float	
PWM Channel 13 Dither Phase	301130	Float	
PWM Channel 14 Dither Phase	301132	Float	
PWM Channel 15 Dither Phase	301134	Float	
PWM Channel 16 Dither Phase	301136	Float	
Supply voltage	302000	Float	
Supply current	302002	Float	
CPU temperature	302004	Float	
Pt100 value	302006	Float	
Analog Input Channel 1 (mA)	302008	Float	
Analog Input Channel 2 (mA)	302010	Float	
Analog Input Channel 3 (mA)	302012	Float	
Analog Input Channel 4 (mA)	302014	Float	
Analog Input Channel 5 (mA)	302016	Float	
Analog Input Channel 6 (mA)	302018	Float	
Analog Input Channel 7 (mA)	302020	Float	
Analog Input Channel 8 (mA)	302022	Float	
Analog Input Channel 9 (mA)	302024	Float	
Analog Input Channel 10 (mA)	302026	Float	
Analog Input Channel 11 (mA)	302028	Float	
Analog Input Channel 12 (mA)	302030	Float	
Analog Input Channel 13 (mA)	302032	Float	
Analog Input Channel 14 (mA)	302034	Float	
Analog Input Channel 15 (mA)	302036	Float	
Analog Input Channel 16 (mA)	302038	Float	

Standard Register Range			
Name	**Register**	**Datatype**	**Note**
Analog Input Channel 1 (V)	302040	Float	
Analog Input Channel 2 (V)	302042	Float	
Analog Input Channel 3 (V)	302044	Float	
Analog Input Channel 4 (V)	302046	Float	
Analog Input voltage flags	302048	Uint16	
Analog Input Channel 1 (Scaled)	302049	Float	
Analog Input Channel 2 (Scaled)	302051	Float	
Analog Input Channel 3 (Scaled)	302053	Float	
Analog Input Channel 4 (Scaled)	302055	Float	
Analog Input Channel 5 (Scaled)	302057	Float	
Analog Input Channel 6 (Scaled)	302059	Float	
Analog Input Channel 7 (Scaled)	302061	Float	
Analog Input Channel 8 (Scaled)	302063	Float	
Analog Input Channel 9 (Scaled)	302065	Float	
Analog Input Channel 10 (Scaled)	302067	Float	
Analog Input Channel 11 (Scaled)	302069	Float	
Analog Input Channel 12 (Scaled)	302071	Float	
Analog Input Channel 13 (Scaled)	302073	Float	
Analog Input Channel 14 (Scaled)	302075	Float	
Analog Input Channel 15 (Scaled)	302077	Float	
Analog Input Channel 16 (Scaled)	302079	Float	
Digital Input Levels	302081	Uint16	
Digital Input 1 Count	302082	Uint32	
Digital Input 2 Count	302084	Uint32	
Digital Input 3 Count	302086	Uint32	
Digital Input 4 Count	302088	Uint32	
Digital Input 1 Frequency	302090	Float	
Digital Input 2 Frequency	302092	Float	
Digital Input 3 Frequency	302094	Float	
Digital Input 4 Frequency	302096	Float	
Encoder 1 Count	302098	Int32	
Encoder 2 Count	302100	Int32	

Standard Register Range			
Name	**Register**	**Datatype**	**Note**
Encoder 1 Frequency	302102	Float	
Encoder 2 Frequency	302104	Float	
Flow 1	302106	Float	
Flow 2	302108	Float	
Flow 3	302110	Float	
Flow 4	302112	Float	
Accumulated Flow 1	302114	Float	
Accumulated Flow 2	302116	Float	
Accumulated Flow 3	302118	Float	
Accumulated Flow 4	302120	Float	
Analog Input Channel 01 range	303000	Float	
Analog Input Channel 02 range	303002	Float	
Analog Input Channel 03 range	303004	Float	
Analog Input Channel 04 range	303006	Float	
Analog Input Channel 05 range	303008	Float	
Analog Input Channel 06 range	303010	Float	
Analog Input Channel 07 range	303012	Float	
Analog Input Channel 08 range	303014	Float	
Analog Input Channel 09 range	303016	Float	
Analog Input Channel 10 range	303018	Float	
Analog Input Channel 11 range	303020	Float	
Analog Input Channel 12 range	303022	Float	
Analog Input Channel 13 range	303024	Float	
Analog Input Channel 14 range	303026	Float	
Analog Input Channel 15 range	303028	Float	
Analog Input Channel 16 range	303030	Float	
Analog Input Channel 01 Gain	303032	Float	
Analog Input Channel 01 Offset	303034	Float	
Analog Input Channel 02 Gain	303036	Float	
Analog Input Channel 02 Offset	303038	Float	
Analog Input Channel 03 Gain	303040	Float	
Analog Input Channel 03 Offset	303042	Float	

Standard Register Range			
Name	**Register**	**Datatype**	**Note**
Analog Input Channel 04 Gain	303044	Float	
Analog Input Channel 04 Offset	303046	Float	
Analog Input Channel 05 Gain	303048	Float	
Analog Input Channel 05 Offset	303050	Float	
Analog Input Channel 06 Gain	303052	Float	
Analog Input Channel 06 Offset	303054	Float	
Analog Input Channel 07 Gain	303056	Float	
Analog Input Channel 07 Offset	303058	Float	
Analog Input Channel 08 Gain	303060	Float	
Analog Input Channel 08 Offset	303062	Float	
Analog Input Channel 09 Gain	303064	Float	
Analog Input Channel 09 Offset	303066	Float	
Analog Input Channel 10 Gain	303068	Float	
Analog Input Channel 10 Offset	303070	Float	
Analog Input Channel 11 Gain	303072	Float	
Analog Input Channel 11 Offset	303074	Float	
Analog Input Channel 12 Gain	303076	Float	
Analog Input Channel 12 Offset	303078	Float	
Analog Input Channel 13 Gain	303080	Float	
Analog Input Channel 13 Offset	303082	Float	
Analog Input Channel 14 Gain	303084	Float	
Analog Input Channel 14 Offset	303086	Float	
Analog Input Channel 15 Gain	303088	Float	
Analog Input Channel 15 Offset	303090	Float	
Analog Input Channel 16 Gain	303092	Float	
Analog Input Channel 16 Offset	303094	Float	
Pt100 Gain	303096	Float	
Pt100 Offset	303098	Float	
Flow 1 k-factor	303100	Float	
Flow 2 k-factor	303102	Float	
Flow 3 k-factor	303104	Float	
Flow 4 k-factor	303106	Float	

Standard Register Range			
Name	**Register**	**Datatype**	**Note**
Analog Input 1 Filter Constant	303108	Float	
Analog Input 2 Filter Constant	303110	Float	
Analog Input 3 Filter Constant	303112	Float	
Analog Input 4 Filter Constant	303114	Float	
Analog Input 5 Filter Constant	303116	Float	
Analog Input 6 Filter Constant	303118	Float	
Analog Input 7 Filter Constant	303120	Float	
Analog Input 8 Filter Constant	303122	Float	
Analog Input 9 Filter Constant	303124	Float	
Analog Input 10 Filter Constant	303126	Float	
Analog Input 11 Filter Constant	303128	Float	
Analog Input 12 Filter Constant	303130	Float	
Analog Input 13 Filter Constant	303132	Float	
Analog Input 14 Filter Constant	303134	Float	
Analog Input 15 Filter Constant	303136	Float	
Analog Input 16 Filter Constant	303138	Float	
Pt100 Filter Constant	303140	Float	
Pt100 Low Limit	303142	Float	
Pt100 High Limit	303144	Float	
Analog Input 1 Raw value	304000	Int16	
Analog Input 2 Raw value	304001	Int16	
Analog Input 3 Raw value	304002	Int16	
Analog Input 4 Raw value	304003	Int16	
Analog Input 5 Raw value	304004	Int16	
Analog Input 6 Raw value	304005	Int16	
Analog Input 7 Raw value	304006	Int16	
Analog Input 8 Raw value	304007	Int16	
Analog Input 9 Raw value	304008	Int16	
Analog Input 10 Raw value	304009	Int16	
Analog Input 11 Raw value	304010	Int16	
Analog Input 12 Raw value	304011	Int16	
Analog Input 13 Raw value	304012	Int16	

Standard Register Range			
Name	**Register**	**Datatype**	**Note**
Analog Input 14 Raw value	304013	Int16	
Analog Input 15 Raw value	304014	Int16	
Analog Input 16 Raw value	304015	Int16	
Analog Input 1 mA Gain	304016	Float	
Analog Input 1 mA Offset	304018	Float	
Analog Input 2 mA Gain	304020	Float	
Analog Input 2 mA Offset	304022	Float	
Analog Input 3 mA Gain	304024	Float	
Analog Input 3 mA Offset	304026	Float	
Analog Input 4 mA Gain	304028	Float	
Analog Input 4 mA Offset	304030	Float	
Analog Input 5 mA Gain	304032	Float	
Analog Input 5 mA Offset	304034	Float	
Analog Input 6 mA Gain	304036	Float	
Analog Input 6 mA Offset	304038	Float	
Analog Input 7 mA Gain	304040	Float	
Analog Input 7 mA Offset	304042	Float	
Analog Input 8 mA Gain	304044	Float	
Analog Input 8 mA Offset	304046	Float	
Analog Input 9 mA Gain	304048	Float	
Analog Input 9 mA Offset	304050	Float	
Analog Input 10 mA Gain	304052	Float	
Analog Input 10 mA Offset	304054	Float	
Analog Input 11 mA Gain	304056	Float	
Analog Input 11 mA Offset	304058	Float	
Analog Input 12 mA Gain	304060	Float	
Analog Input 12 mA Offset	304062	Float	
Analog Input 13 mA Gain	304064	Float	
Analog Input 13 mA Offset	304066	Float	
Analog Input 14 mA Gain	304068	Float	
Analog Input 14 mA Offset	304070	Float	
Analog Input 15 mA Gain	304072	Float	

Standard Register Range			
Name	**Register**	**Datatype**	**Note**
Analog Input 15 mA Offset	304074	Float	
Analog Input 16 mA Gain	304076	Float	
Analog Input 16 mA Offset	304078	Float	
Analog Input 1 V Gain	304080	Float	
Analog Input 1 V Offset	304082	Float	
Analog Input 2 V Gain	304084	Float	
Analog Input 2 V Offset	304086	Float	
Analog Input 3 V Gain	304088	Float	
Analog Input 3 V Offset	304090	Float	
Analog Input 4 V Gain	304092	Float	
Analog Input 4 V Offset	304094	Float	
Supply voltage Gain	304096	Float	
Supply voltage Offset	304098	Float	
Supply current Gain	304100	Float	
Supply current Offset	304102	Float	
CPU Temperature Gain	304104	Float	
CPU Temperature Offset	304106	Float	
Raw Water Alarm value	304108	Uint16	
WA pin time	304109	Uint32	
WA wait time	304111	Uint32	
WA Low Limit	304113	Uint16	
WA High Limit	304114	Uint16	
Supply Voltage Low Limit	304115	Float	
Supply Voltage High Limit	304117	Float	
Supply Current Low Limit	304119	Float	
Supply Current High Limit	304121	Float	
CPU Temperature Low Limit	304123	Float	
CPU Temperature High Limit	304125	Float	
Supply Voltage Filter Constant	304127	Float	
Supply Current Filter Constant	304129	Float	
Pt100 Filter Constant	304131	Float	
CPU Temperature Filter Constant	304133	Float	

Standard Register Range			
Name	**Register**	**Datatype**	**Note**
Ipv4 Address	400050	Uint32	
Subnet Mask	400052	Uint32	
Default Gateway	400054	Uint32	
MAC address[0]	400056	Uint16	
MAC address[1]	400057	Uint16	
MAC address[2]	400058	Uint16	
Baudrate RS232	400059	Uint32	
Baudrate RS485	400061	Uint32	
Modbus Slave Base Address	400063	Uint8	
Lost-coms timeout	400065	Uint16	
Lost-coms Keep Value	400066	Uint16	
Power-on default output 1	400067	Float	
Power-on default output 2	400069	Float	
Power-on default output 3	400071	Float	
Power-on default output 4	400073	Float	
Power-on default output 5	400075	Float	
Power-on default output 6	400077	Float	
Power-on default output 7	400079	Float	
Power-on default output 8	400081	Float	
Power-on default output 9	400083	Float	
Power-on default output 10	400085	Float	
Power-on default output 11	400087	Float	
Power-on default output 12	400089	Float	
Power-on default output 13	400091	Float	
Power-on default output 14	400093	Float	
Power-on default output 15	400095	Float	
Power-on default output 16	400097	Float	
Sensor Power configuration	400099	Uint16	
Reset System Warnings	400102	Uint16	Always reads 0
Reset Sensor Alarms	400103	Uint16	Always reads 0
Reset PWM Faults	400104	Uint16	Always reads 0
Sensor Power Enable	400111	Uint8	

Standard Register Range			
Name	**Register**	**Datatype**	**Note**
Restart command	400112	Uint8	
EEPROM flush command	400113	Uint8	
EEPROM clean-up command	400114	Uint8	
rs232_terminal	400115	Uint8	
rs485_terminal	400116	Uint8	
Lost COM's output 1	400117	Float	From version 1.3.0
Lost COM's output 2	400119	Float	From version 1.3.0
Lost COM's output 3	400121	Float	From version 1.3.0
Lost COM's output 4	400123	Float	From version 1.3.0
Lost COM's output 5	400125	Float	From version 1.3.0
Lost COM's output 6	400127	Float	From version 1.3.0
Lost COM's output 7	400129	Float	From version 1.3.0
Lost COM's output 8	400131	Float	From version 1.3.0
Lost COM's output 9	400133	Float	From version 1.3.0
Lost COM's output 10	400135	Float	From version 1.3.0
Lost COM's output 11	400137	Float	From version 1.3.0
Lost COM's output 12	400139	Float	From version 1.3.0
Lost COM's output 13	400141	Float	From version 1.3.0
Lost COM's output 14	400143	Float	From version 1.3.0
Lost COM's output 15	400145	Float	From version 1.3.0
Lost COM's output 16	400147	Float	From version 1.3.0
Modbus Gateway Enabled	400149	Bool	From version 1.4.0
PWM Channel 1 Duty	401000	Float	
PWM Channel 2 Duty	401002	Float	
PWM Channel 3 Duty	401004	Float	
PWM Channel 4 Duty	401006	Float	
PWM Channel 5 Duty	401008	Float	
PWM Channel 6 Duty	401010	Float	
PWM Channel 7 Duty	401012	Float	
PWM Channel 8 Duty	401014	Float	
PWM Channel 9 Duty	401016	Float	
PWM Channel 10 Duty	401018	Float	

Standard Register Range			
Name	**Register**	**Datatype**	**Note**
PWM Channel 11 Duty	401020	Float	
PWM Channel 12 Duty	401022	Float	
PWM Channel 13 Duty	401024	Float	
PWM Channel 14 Duty	401026	Float	
PWM Channel 15 Duty	401028	Float	
PWM Channel 16 Duty	401030	Float	
PWM synchronization command	401032	Uint8	
PWM Ignore faults	401033	Uint16	
PWM Frequency Channels 1-4	401034	Float	
PWM Frequency Channels 5-8	401036	Float	
PWM Frequency Channels 9-12	401038	Float	
PWM Frequency Channels 13-16	401040	Float	
PWM Channel 1 Dither Duty	401042	Float	
PWM Channel 2 Dither Duty	401044	Float	
PWM Channel 3 Dither Duty	401046	Float	
PWM Channel 4 Dither Duty	401048	Float	
PWM Channel 5 Dither Duty	401050	Float	
PWM Channel 6 Dither Duty	401052	Float	
PWM Channel 7 Dither Duty	401054	Float	
PWM Channel 8 Dither Duty	401056	Float	
PWM Channel 9 Dither Duty	401058	Float	
PWM Channel 10 Dither Duty	401060	Float	
PWM Channel 11 Dither Duty	401062	Float	
PWM Channel 12 Dither Duty	401064	Float	
PWM Channel 13 Dither Duty	401066	Float	
PWM Channel 14 Dither Duty	401068	Float	
PWM Channel 15 Dither Duty	401070	Float	
PWM Channel 16 Dither Duty	401072	Float	
PWM Channel 1 Dither Frequency	401074	Float	
PWM Channel 2 Dither Frequency	401076	Float	
PWM Channel 3 Dither Frequency	401078	Float	
PWM Channel 4 Dither Frequency	401080	Float	

Standard Register Range			
Name	**Register**	**Datatype**	**Note**
PWM Channel 5 Dither Frequency	401082	Float	
PWM Channel 6 Dither Frequency	401084	Float	
PWM Channel 7 Dither Frequency	401086	Float	
PWM Channel 8 Dither Frequency	401088	Float	
PWM Channel 9 Dither Frequency	401090	Float	
PWM Channel 10 Dither Frequency	401092	Float	
PWM Channel 11 Dither Frequency	401094	Float	
PWM Channel 12 Dither Frequency	401096	Float	
PWM Channel 13 Dither Frequency	401098	Float	
PWM Channel 14 Dither Frequency	401100	Float	
PWM Channel 15 Dither Frequency	401102	Float	
PWM Channel 16 Dither Frequency	401104	Float	
PWM Channel 1 Dither Phase	401106	Float	
PWM Channel 2 Dither Phase	401108	Float	
PWM Channel 3 Dither Phase	401110	Float	
PWM Channel 4 Dither Phase	401112	Float	
PWM Channel 5 Dither Phase	401114	Float	
PWM Channel 6 Dither Phase	401116	Float	
PWM Channel 7 Dither Phase	401118	Float	
PWM Channel 8 Dither Phase	401120	Float	
PWM Channel 9 Dither Phase	401122	Float	
PWM Channel 10 Dither Phase	401124	Float	
PWM Channel 11 Dither Phase	401126	Float	
PWM Channel 12 Dither Phase	401128	Float	
PWM Channel 13 Dither Phase	401130	Float	
PWM Channel 14 Dither Phase	401132	Float	
PWM Channel 15 Dither Phase	401134	Float	
PWM Channel 16 Dither Phase	401136	Float	
Reset digital input 1 count	402082	Uint8	
Reset digital input 2 count	402084	Uint8	
Reset digital input 3 count	402086	Uint8	
Reset digital input 4 count	402088	Uint8	

Standard Register Range			
Name	**Register**	**Datatype**	**Note**
Reset encoder 1 count	402098	Uint8	
Reset encoder 2 count	402100	Uint8	
Analog Input Channel 01 range	403000	Float	
Analog Input Channel 02 range	403002	Float	
Analog Input Channel 03 range	403004	Float	
Analog Input Channel 04 range	403006	Float	
Analog Input Channel 05 range	403008	Float	
Analog Input Channel 06 range	403010	Float	
Analog Input Channel 07 range	403012	Float	
Analog Input Channel 08 range	403014	Float	
Analog Input Channel 09 range	403016	Float	
Analog Input Channel 10 range	403018	Float	
Analog Input Channel 11 range	403020	Float	
Analog Input Channel 12 range	403022	Float	
Analog Input Channel 13 range	403024	Float	
Analog Input Channel 14 range	403026	Float	
Analog Input Channel 15 range	403028	Float	
Analog Input Channel 16 range	403030	Float	
Analog Input Channel 01 Gain	403032	Float	
Analog Input Channel 01 Offset	403034	Float	
Analog Input Channel 02 Gain	403036	Float	
Analog Input Channel 02 Offset	403038	Float	
Analog Input Channel 03 Gain	403040	Float	
Analog Input Channel 03 Offset	403042	Float	
Analog Input Channel 04 Gain	403044	Float	
Analog Input Channel 04 Offset	403046	Float	
Analog Input Channel 05 Gain	403048	Float	
Analog Input Channel 05 Offset	403050	Float	
Analog Input Channel 06 Gain	403052	Float	
Analog Input Channel 06 Offset	403054	Float	
Analog Input Channel 07 Gain	403056	Float	
Analog Input Channel 07 Offset	403058	Float	

Standard Register Range			
Name	**Register**	**Datatype**	**Note**
Analog Input Channel 08 Gain	403060	Float	
Analog Input Channel 08 Offset	403062	Float	
Analog Input Channel 09 Gain	403064	Float	
Analog Input Channel 09 Offset	403066	Float	
Analog Input Channel 10 Gain	403068	Float	
Analog Input Channel 10 Offset	403070	Float	
Analog Input Channel 11 Gain	403072	Float	
Analog Input Channel 11 Offset	403074	Float	
Analog Input Channel 12 Gain	403076	Float	
Analog Input Channel 12 Offset	403078	Float	
Analog Input Channel 13 Gain	403080	Float	
Analog Input Channel 13 Offset	403082	Float	
Analog Input Channel 14 Gain	403084	Float	
Analog Input Channel 14 Offset	403086	Float	
Analog Input Channel 15 Gain	403088	Float	
Analog Input Channel 15 Offset	403090	Float	
Analog Input Channel 16 Gain	403092	Float	
Analog Input Channel 16 Offset	403094	Float	
Pt100 Gain	403096	Float	
Pt100 Offset	403098	Float	
Flow 1 k-factor	403100	Float	
Flow 2 k-factor	403102	Float	
Flow 3 k-factor	403104	Float	
Flow 4 k-factor	403106	Float	
Analog Input 1 Filter Constant	403108	Float	
Analog Input 2 Filter Constant	403110	Float	
Analog Input 3 Filter Constant	403112	Float	
Analog Input 4 Filter Constant	403114	Float	
Analog Input 5 Filter Constant	403116	Float	
Analog Input 6 Filter Constant	403118	Float	
Analog Input 7 Filter Constant	403120	Float	
Analog Input 8 Filter Constant	403122	Float	

Standard Register Range			
Name	**Register**	**Datatype**	**Note**
Analog Input 9 Filter Constant	403124	Float	
Analog Input 10 Filter Constant	403126	Float	
Analog Input 11 Filter Constant	403128	Float	
Analog Input 12 Filter Constant	403130	Float	
Analog Input 13 Filter Constant	403132	Float	
Analog Input 14 Filter Constant	403134	Float	
Analog Input 15 Filter Constant	403136	Float	
Analog Input 16 Filter Constant	403138	Float	
Pt100 Filter Constant	403140	Float	
Pt100 Low Limit	403142	Float	
Pt100 High Limit	403144	Float	

10.2 Table of registers, concentrated ranges

For a more efficient communication, a selection of the most relevant registers has been duplicated to a set of new register ranges. In these ranges the addresses are sequential without gaps, which allows for fewer Modbus request to read or write the relevant registers.

Note: The concentrated register ranges are available from firmware version 1.5.3.

There are two ranges. Range 1 includes the analogue inputs in mA values, while range 2 includes the analogue inputs in scaled values. The two ranges share the write sections. The ranges have four sections; operational values read/write sections and configuration values read/write sections. The operational sections include the input, output, onboard sensors, warning, and alarm registers. The configuration sections include the device identification and configuration registers. The operational registers are dynamic and should be read regularly for monitoring. The configuration registers are static and can be read once on connect or on demand.

	Range 1 (analogue inputs: mA)	Range 2 (analogue inputs: scaled)
Read Operational	310000 to 310133	311000 to 311133
Read Configuration	310135 to 310374	311135 to 311374
Write Operational	410000 to 410043	
Write Configuration	410044 to 410268	

10.2.1 Read Operational Values

Concentrated Register Range – Read Operational					
Name	**Original Register**	**Datatype**	**Range 1 (mA)**	**Range 2 (Scaled)**	**Note**
# PWM Outputs #	---	---	---	---	
PWM Channel 1 Duty	301000	Float	310000	311000	
PWM Channel 2 Duty	301002	Float	310002	311002	
PWM Channel 3 Duty	301004	Float	310004	311004	
PWM Channel 4 Duty	301006	Float	310006	311006	
PWM Channel 5 Duty	301008	Float	310008	311008	
PWM Channel 6 Duty	301010	Float	310010	311010	
PWM Channel 7 Duty	301012	Float	310012	311012	
PWM Channel 8 Duty	301014	Float	310014	311014	
PWM Channel 9 Duty	301016	Float	310016	311016	
PWM Channel 10 Duty	301018	Float	310018	311018	
PWM Channel 11 Duty	301020	Float	310020	311020	
PWM Channel 12 Duty	301022	Float	310022	311022	
PWM Channel 13 Duty	301024	Float	310024	311024	
PWM Channel 14 Duty	301026	Float	310026	311026	

Concentrated Register Range – Read Operational					
Name	**Original Register**	**Datatype**	**Range 1 (mA)**	**Range 2 (Scaled)**	**Note**
PWM Channel 15 Duty	301028	Float	310028	311028	
PWM Channel 16 Duty	301030	Float	310030	311030	
# Sensor Power #	---	---	---	---	
Sensor Power Enable	300111	Uint8	310032	311032	
# PIO36 Status #	---	---	---	---	
Heartbeat	300014	Uint16	310033	311033	
System Warnings	300102	Uint16	310034	311034	
Sensor Alarms	300103	Uint16	310035	311035	
PWM Faults	300104	Uint16	310036	311036	
# Internal Sensors #	---	---	---	---	
Supply voltage	302000	Float	310037	311037	
Supply current	302002	Float	310039	311039	
CPU temperature	302004	Float	310041	311041	
Pt100 value	302006	Float	310043	311043	
# Analogue Input #	---	---	---	---	
Analog Input Channel 1 (mA)	302008	Float	310045	-	
Analog Input Channel 2 (mA)	302010	Float	310047	-	
Analog Input Channel 3 (mA)	302012	Float	310049	-	
Analog Input Channel 4 (mA)	302014	Float	310051	-	
Analog Input Channel 5 (mA)	302016	Float	310053	-	
Analog Input Channel 6 (mA)	302018	Float	310055	-	
Analog Input Channel 7 (mA)	302020	Float	310057	-	
Analog Input Channel 8 (mA)	302022	Float	310059	-	
Analog Input Channel 9 (mA)	302024	Float	310061	-	
Analog Input Channel 10 (mA)	302026	Float	310063	-	

Concentrated Register Range – Read Operational					
Name	**Original Register**	**Datatype**	**Range 1 (mA)**	**Range 2 (Scaled)**	**Note**
Analog Input Channel 11 (mA)	302028	Float	310065	-	
Analog Input Channel 12 (mA)	302030	Float	310067	-	
Analog Input Channel 13 (mA)	302032	Float	310069	-	
Analog Input Channel 14 (mA)	302034	Float	310071	-	
Analog Input Channel 15 (mA)	302036	Float	310073	-	
Analog Input Channel 16 (mA)	302038	Float	310075	-	
Analog Input Channel 1 (Scaled)	302049	Float	-	311045	
Analog Input Channel 2 (Scaled)	302051	Float	-	311047	
Analog Input Channel 3 (Scaled)	302053	Float	-	311049	
Analog Input Channel 4 (Scaled)	302055	Float	-	311051	
Analog Input Channel 5 (Scaled)	302057	Float	-	311053	
Analog Input Channel 6 (Scaled)	302059	Float	-	311055	
Analog Input Channel 7 (Scaled)	302061	Float	-	311057	
Analog Input Channel 8 (Scaled)	302063	Float	-	311059	
Analog Input Channel 9 (Scaled)	302065	Float	-	311061	
Analog Input Channel 10 (Scaled)	302067	Float	-	311063	
Analog Input Channel 11 (Scaled)	302069	Float	-	311065	
Analog Input Channel 12 (Scaled)	302071	Float	-	311067	
Analog Input Channel 13 (Scaled)	302073	Float	-	311069	

Concentrated Register Range – Read Operational					
Name	**Original Register**	**Datatype**	**Range 1 (mA)**	**Range 2 (Scaled)**	**Note**
Analog Input Channel 14 (Scaled)	302075	Float	-	311071	
Analog Input Channel 15 (Scaled)	302077	Float	-	311073	
Analog Input Channel 16 (Scaled)	302079	Float	-	311075	
Analog Input Channel 1 (V)	302040	Float	310077	311077	
Analog Input Channel 2 (V)	302042	Float	310079	311079	
Analog Input Channel 3 (V)	302044	Float	310081	311081	
Analog Input Channel 4 (V)	302046	Float	310083	311083	
Analog Input voltage flags	302048	Uint16	310085	311085	
# Digital Input #	---	---	---	---	
Digital Input Levels	302081	Uint16	310086	311086	
Digital Input 1 Count	302082	Uint32	310087	311087	
Digital Input 2 Count	302084	Uint32	310089	311089	
Digital Input 3 Count	302086	Uint32	310091	311091	
Digital Input 4 Count	302088	Uint32	310093	311093	
Digital Input 1 Frequency	302090	Float	310095	311095	
Digital Input 2 Frequency	302092	Float	310097	311097	
Digital Input 3 Frequency	302094	Float	310099	311099	
Digital Input 4 Frequency	302096	Float	310101	311101	
Flow 1	302106	Float	310103	311103	
Flow 2	302108	Float	310105	311105	
Flow 3	302110	Float	310107	311107	
Flow 4	302112	Float	310109	311109	
Accumulated Flow 1	302114	Float	310111	311111	
Accumulated Flow 2	302116	Float	310113	311113	
Accumulated Flow 3	302118	Float	310115	311115	
Accumulated Flow 4	302120	Float	310117	311117	
Encoder 1 Count	302098	Int32	310119	311119	
Encoder 2 Count	302100	Int32	310121	311121	
Encoder 1 Frequency	302102	Float	310123	311123	

Concentrated Register Range – Read Operational					
Name	**Original Register**	**Datatype**	**Range 1 (mA)**	**Range 2 (Scaled)**	**Note**
Encoder 2 Frequency	302104	Float	310125	311125	
# Diagnostics #	---	---	---	---	
Uptime	300100	Float	310127	311127	
Modbus Packet Count	300105	Uint32	310129	311129	
Modbus Dropped Packets Count	300107	Uint32	310131	311131	
EEPROM write counter	300109	Uint32	310133	311133	

10.2.2 Read Configuration

Concentrated Register Range – Read Configuration					
Name	**Original Register**	**Datatype**	**Range 1 (mA)**	**Range 2 (Scaled)**	**Note**
# Identification #	---	---	---	---	
Product Number	300001	Uint32	310135	311135	
Serial Number	300003	Uint32	310137	311137	
Hardware Version	300005	Uint32	310139	311139	
FW version Major	300007	Uint16	310141	311141	
FW version Minor	300008	Uint16	310142	311142	
FW version Patch	300009	Uint16	310143	311143	
Firmware CRC	300010	Uint32	310144	311144	
Configuration CRC	300012	Uint32	310146	311146	
# PWM Output Configuration #	---	---	---	---	
Lost-coms Keep Value	300066	Uint16	310148	311148	
Lost COM's output 1	300117	Float	310149	311149	
Lost COM's output 2	300119	Float	310151	311151	
Lost COM's output 3	300121	Float	310153	311153	
Lost COM's output 4	300123	Float	310155	311155	
Lost COM's output 5	300125	Float	310157	311157	
Lost COM's output 6	300127	Float	310159	311159	
Lost COM's output 7	300129	Float	310161	311161	
Lost COM's output 8	300131	Float	310163	311163	
Lost COM's output 9	300133	Float	310165	311165	

Concentrated Register Range – Read Configuration					
Name	**Original Register**	**Datatype**	**Range 1 (mA)**	**Range 2 (Scaled)**	**Note**
Lost COM's output 10	300135	Float	310167	311167	
Lost COM's output 11	300137	Float	310169	311169	
Lost COM's output 12	300139	Float	310171	311171	
Lost COM's output 13	300141	Float	310173	311173	
Lost COM's output 14	300143	Float	310175	311175	
Lost COM's output 15	300145	Float	310177	311177	
Lost COM's output 16	300147	Float	310179	311179	
Power-on default output 1	300067	Float	310181	311181	
Power-on default output 2	300069	Float	310183	311183	
Power-on default output 3	300071	Float	310185	311185	
Power-on default output 4	300073	Float	310187	311187	
Power-on default output 5	300075	Float	310189	311189	
Power-on default output 6	300077	Float	310191	311191	
Power-on default output 7	300079	Float	310193	311193	
Power-on default output 8	300081	Float	310195	311195	
Power-on default output 9	300083	Float	310197	311197	
Power-on default output 10	300085	Float	310199	311199	
Power-on default output 11	300087	Float	310201	311201	
Power-on default output 12	300089	Float	310203	311203	
Power-on default output 13	300091	Float	310205	311205	
Power-on default output 14	300093	Float	310207	311207	
Power-on default output 15	300095	Float	310209	311209	
Power-on default output 16	300097	Float	310211	311211	
PWM Ignore Faults	301033	Uint16	310213	311213	
# Sensor Power #	---	---	---	---	
Sensor power configuration	300099	Uint16	310214	311214	
# Communication #	---	---	---	---	
Ipv4 Address	300050	IPv4 Address	310215	311215	
Subnet Mask	300052	IPv4 Address	310217	311217	

Concentrated Register Range – Read Configuration					
Name	**Original Register**	**Datatype**	**Range 1 (mA)**	**Range 2 (Scaled)**	**Note**
Default Gateway	300054	IPv4 Address	310219	311219	
Ethernet Port 1 Config	-	Uint8	310221	311221	
Ethernet Port 2 Config	-	Uint8	310222	311222	
RSTP enabled	-	Uint8	310223	311223	
Modbus gateway enabled	300149	Bool	310224	311224	
Baudrate RS232	300059	Uint32	310225	311225	
Baudrate RS485	300061	Uint32	310227	311227	
Modbus base address	300063	Uint8	310229	311229	
Modbus address	300064	Uint8	310230	311230	
Lost-coms timeout	300065	Uint16	310231	311231	
# Sensor Configuration #	---	---	---	---	
Pt100 Gain	303096	Float	310232	311232	
Pt100 Offset	303098	Float	310234	311234	
Pt100 Low Limit	303142	Float	310236	311236	
Pt100 High Limit	303144	Float	310238	311238	
Flow 1 k-factor	303100	Float	310240	311240	
Flow 2 k-factor	303102	Float	310242	311242	
Flow 3 k-factor	303104	Float	310244	311244	
Flow 4 k-factor	303106	Float	310246	311246	
Analog Input 1 Filter Constant	303108	Float	310248	311248	
Analog Input 2 Filter Constant	303110	Float	310250	311250	
Analog Input 3 Filter Constant	303112	Float	310252	311252	
Analog Input 4 Filter Constant	303114	Float	310254	311254	
Analog Input 5 Filter Constant	303116	Float	310256	311256	
Analog Input 6 Filter Constant	303118	Float	310258	311258	
Analog Input 7 Filter Constant	303120	Float	310260	311260	

Concentrated Register Range – Read Configuration					
Name	**Original Register**	**Datatype**	**Range 1 (mA)**	**Range 2 (Scaled)**	**Note**
Analog Input 8 Filter Constant	303122	Float	310262	311262	
Analog Input 9 Filter Constant	303124	Float	310264	311264	
Analog Input 10 Filter Constant	303126	Float	310266	311266	
Analog Input 11 Filter Constant	303128	Float	310268	311268	
Analog Input 12 Filter Constant	303130	Float	310270	311270	
Analog Input 13 Filter Constant	303132	Float	310272	311272	
Analog Input 14 Filter Constant	303134	Float	310274	311274	
Analog Input 15 Filter Constant	303136	Float	310276	311276	
Analog Input 16 Filter Constant	303138	Float	310278	311278	
Analog Input Channel 01 range	303000	Float	310280	311280	From version 1.5.4
Analog Input Channel 02 range	303002	Float	310282	311282	From version 1.5.4
Analog Input Channel 03 range	303004	Float	310284	311284	From version 1.5.4
Analog Input Channel 04 range	303006	Float	310286	311286	From version 1.5.4
Analog Input Channel 05 range	303008	Float	310288	311288	From version 1.5.4
Analog Input Channel 06 range	303010	Float	310290	311290	From version 1.5.4
Analog Input Channel 07 range	303012	Float	310292	311292	From version 1.5.4
Analog Input Channel 08 range	303014	Float	310294	311294	From version 1.5.4
Analog Input Channel 09 range	303016	Float	310296	311296	From version 1.5.4
Analog Input Channel 10 range	303018	Float	310298	311298	From version 1.5.4

Concentrated Register Range – Read Configuration					
Name	**Original Register**	**Datatype**	**Range 1 (mA)**	**Range 2 (Scaled)**	**Note**
Analog Input Channel 11 range	303020	Float	310300	311300	From version 1.5.4
Analog Input Channel 12 range	303022	Float	310302	311302	From version 1.5.4
Analog Input Channel 13 range	303024	Float	310304	311304	From version 1.5.4
Analog Input Channel 14 range	303026	Float	310306	311306	From version 1.5.4
Analog Input Channel 15 range	303028	Float	310308	311308	From version 1.5.4
Analog Input Channel 16 range	303030	Float	310310	311310	From version 1.5.4
Analog Input Channel 01 Gain	303032	Float	310312	311312	From version 1.5.4
Analog Input Channel 01 Offset	303034	Float	310314	311314	From version 1.5.4
Analog Input Channel 02 Gain	303036	Float	310316	311316	From version 1.5.4
Analog Input Channel 02 Offset	303038	Float	310318	311318	From version 1.5.4
Analog Input Channel 03 Gain	303040	Float	310320	311320	From version 1.5.4
Analog Input Channel 03 Offset	303042	Float	310322	311322	From version 1.5.4
Analog Input Channel 04 Gain	303044	Float	310324	311324	From version 1.5.4
Analog Input Channel 04 Offset	303046	Float	310326	311326	From version 1.5.4
Analog Input Channel 05 Gain	303048	Float	310328	311328	From version 1.5.4
Analog Input Channel 05 Offset	303050	Float	310330	311330	From version 1.5.4
Analog Input Channel 06 Gain	303052	Float	310332	311332	From version 1.5.4
Analog Input Channel 06 Offset	303054	Float	310334	311334	From version 1.5.4
Analog Input Channel 07 Gain	303056	Float	310336	311336	From version 1.5.4

Concentrated Register Range – Read Configuration					
Name	**Original Register**	**Datatype**	**Range 1 (mA)**	**Range 2 (Scaled)**	**Note**
Analog Input Channel 07 Offset	303058	Float	310338	311338	From version 1.5.4
Analog Input Channel 08 Gain	303060	Float	310340	311340	From version 1.5.4
Analog Input Channel 08 Offset	303062	Float	310342	311342	From version 1.5.4
Analog Input Channel 09 Gain	303064	Float	310344	311344	From version 1.5.4
Analog Input Channel 09 Offset	303066	Float	310346	311346	From version 1.5.4
Analog Input Channel 10 Gain	303068	Float	310348	311348	From version 1.5.4
Analog Input Channel 10 Offset	303070	Float	310350	311350	From version 1.5.4
Analog Input Channel 11 Gain	303072	Float	310352	311352	From version 1.5.4
Analog Input Channel 11 Offset	303074	Float	310354	311354	From version 1.5.4
Analog Input Channel 12 Gain	303076	Float	310356	311356	From version 1.5.4
Analog Input Channel 12 Offset	303078	Float	310358	311358	From version 1.5.4
Analog Input Channel 13 Gain	303080	Float	310360	311360	From version 1.5.4
Analog Input Channel 13 Offset	303082	Float	310362	311362	From version 1.5.4
Analog Input Channel 14 Gain	303084	Float	310364	311364	From version 1.5.4
Analog Input Channel 14 Offset	303086	Float	310366	311366	From version 1.5.4
Analog Input Channel 15 Gain	303088	Float	310368	311368	From version 1.5.4
Analog Input Channel 15 Offset	303090	Float	310370	311370	From version 1.5.4
Analog Input Channel 16 Gain	303092	Float	310372	311372	From version 1.5.4
Analog Input Channel 16 Offset	303094	Float	310374	311374	From version 1.5.4

10.2.3 Write Operational Values

Concentrated Register Range – Write Operational				
Name	**Original Register**	**Datatype**	**Range 1/2**	**Note**
# PWM Output #	---	---	---	
PWM Channel 1 Duty	401000	Float	410000	
PWM Channel 2 Duty	401002	Float	410002	
PWM Channel 3 Duty	401004	Float	410004	
PWM Channel 4 Duty	401006	Float	410006	
PWM Channel 5 Duty	401008	Float	410008	
PWM Channel 6 Duty	401010	Float	410010	
PWM Channel 7 Duty	401012	Float	410012	
PWM Channel 8 Duty	401014	Float	410014	
PWM Channel 9 Duty	401016	Float	410016	
PWM Channel 10 Duty	401018	Float	410018	
PWM Channel 11 Duty	401020	Float	410020	
PWM Channel 12 Duty	401022	Float	410022	
PWM Channel 13 Duty	401024	Float	410024	
PWM Channel 14 Duty	401026	Float	410026	
PWM Channel 15 Duty	401028	Float	410028	
PWM Channel 16 Duty	401030	Float	410030	
# Sensor Power #	---	---	---	
Sensor Power Enable	400111	Uint8	410032	
# Reset				
Reset System warnings	400102	Uint16	410033	Always reads 0
Reset Sensor alarms	400103	Uint16	410034	Always reads 0
Reset PWM Faults	400104	Uint16	410035	Always reads 0
Restart command	400112	Uint8	410036	
EEPROM flush command	400113	Uint8	410037	
Reset digital input 1 count	402082	Uint8	410038	
Reset digital input 2 count	402084	Uint8	410039	
Reset digital input 3 count	402086	Uint8	410040	
Reset digital input 4 count	402088	Uint8	410041	
Reset encoder 1 count	402098	Uint8	410042	

Concentrated Register Range – Write Operational				
Name	**Original Register**	**Datatype**	**Range 1/2**	**Note**
Reset encoder 2 count	402100	Uint8	410043	

10.2.4 Write Configuration

Concentrated Register Range – Write Configuration				
Name	**Original Register**	**Datatype**	**Range 1/2**	**Note**
Lost-coms keep values	400066	Uint16	410044	
Lost COM's output 1	400117	Float	410045	
Lost COM's output 2	400119	Float	410047	
Lost COM's output 3	400121	Float	410049	
Lost COM's output 4	400123	Float	410051	
Lost COM's output 5	400125	Float	410053	
Lost COM's output 6	400127	Float	410055	
Lost COM's output 7	400129	Float	410057	
Lost COM's output 8	400131	Float	410059	
Lost COM's output 9	400133	Float	410061	
Lost COM's output 10	400135	Float	410063	
Lost COM's output 11	400137	Float	410065	
Lost COM's output 12	400139	Float	410067	
Lost COM's output 13	400141	Float	410069	
Lost COM's output 14	400143	Float	410071	
Lost COM's output 15	400145	Float	410073	
Lost COM's output 16	400147	Float	410075	
Power-on default output 1	400067	Float	410077	
Power-on default output 2	400069	Float	410079	
Power-on default output 3	400071	Float	410081	
Power-on default output 4	400073	Float	410083	
Power-on default output 5	400075	Float	410085	
Power-on default output 6	400077	Float	410087	
Power-on default output 7	400079	Float	410089	
Power-on default output 8	400081	Float	410091	

Concentrated Register Range – Write Configuration				
Name	**Original Register**	**Datatype**	**Range 1/2**	**Note**
Power-on default output 9	400083	Float	410093	
Power-on default output 10	400085	Float	410095	
Power-on default output 11	400087	Float	410097	
Power-on default output 12	400089	Float	410099	
Power-on default output 13	400091	Float	410101	
Power-on default output 14	400093	Float	410103	
Power-on default output 15	400095	Float	410105	
Power-on default output 16	400097	Float	410107	
PWM Ignore Faults	401033	Uint16	410109	
Ipv4 Address	400050	Uint32	410110	
Subnet Mask	400052	Uint32	410112	
Default Gateway	400054	Uint32	410114	
Ethernet Port 1 Config	-	Uint8	410116	
Ethernet Port 2 Config	-	Uint8	410117	
RSTP enabled	-	Uint8	410118	
Modbus gateway enabled	400149	Bool	410119	
Baudrate RS242	400059	Uint32	410120	
Baudrate RS485	400061	Uint32	410122	
Modbus slave base address	400063	Uint8	410124	
Lost-coms timeout	400065	Uint16	410125	
Pt100 Gain	403096	Float	410126	
Pt100 Offset	403098	Float	410128	
Pt100 Low Limit	403142	Float	410130	
Pt100 High Limit	403144	Float	410132	
Flow 1 k-factor	403100	Float	410134	
Flow 2 k-factor	403102	Float	410136	
Flow 3 k-factor	403104	Float	410138	
Flow 4 k-factor	403106	Float	410140	
Analog Input 1 Filter Constant	403108	Float	410142	
Analog Input 2 Filter Constant	403110	Float	410144	
Analog Input 3 Filter Constant	403112	Float	410146	

Concentrated Register Range – Write Configuration				
Name	**Original Register**	**Datatype**	**Range 1/2**	**Note**
Analog Input 4 Filter Constant	403114	Float	410148	
Analog Input 5 Filter Constant	403116	Float	410150	
Analog Input 6 Filter Constant	403118	Float	410152	
Analog Input 7 Filter Constant	403120	Float	410154	
Analog Input 8 Filter Constant	403122	Float	410156	
Analog Input 9 Filter Constant	403124	Float	410158	
Analog Input 10 Filter Constant	403126	Float	410160	
Analog Input 11 Filter Constant	403128	Float	410162	
Analog Input 12 Filter Constant	403130	Float	410164	
Analog Input 13 Filter Constant	403132	Float	410166	
Analog Input 14 Filter Constant	403134	Float	410168	
Analog Input 15 Filter Constant	403136	Float	410170	
Analog Input 16 Filter Constant	403138	Float	410172	
Analog Input Channel 01 range	403000	Float	410174	
Analog Input Channel 02 range	403002	Float	410176	
Analog Input Channel 03 range	403004	Float	410178	
Analog Input Channel 04 range	403006	Float	410180	
Analog Input Channel 05 range	403008	Float	410182	
Analog Input Channel 06 range	403010	Float	410184	
Analog Input Channel 07 range	403012	Float	410186	
Analog Input Channel 08 range	403014	Float	410188	
Analog Input Channel 09 range	403016	Float	410190	
Analog Input Channel 10 range	403018	Float	410192	
Analog Input Channel 11 range	403020	Float	410194	
Analog Input Channel 12 range	403022	Float	410196	
Analog Input Channel 13 range	403024	Float	410198	
Analog Input Channel 14 range	403026	Float	410200	
Analog Input Channel 15 range	403028	Float	410202	
Analog Input Channel 16 range	403030	Float	410204	
Analog Input Channel 01 Gain	403032	Float	410206	
Analog Input Channel 01 Offset	403034	Float	410208	

Concentrated Register Range – Write Configuration				
Name	**Original Register**	**Datatype**	**Range 1/2**	**Note**
Analog Input Channel 02 Gain	403036	Float	410210	
Analog Input Channel 02 Offset	403038	Float	410212	
Analog Input Channel 03 Gain	403040	Float	410214	
Analog Input Channel 03 Offset	403042	Float	410216	
Analog Input Channel 04 Gain	403044	Float	410218	
Analog Input Channel 04 Offset	403046	Float	410220	
Analog Input Channel 05 Gain	403048	Float	410222	
Analog Input Channel 05 Offset	403050	Float	410224	
Analog Input Channel 06 Gain	403052	Float	410226	
Analog Input Channel 06 Offset	403054	Float	410228	
Analog Input Channel 07 Gain	403056	Float	410230	
Analog Input Channel 07 Offset	403058	Float	410232	
Analog Input Channel 08 Gain	403060	Float	410234	
Analog Input Channel 08 Offset	403062	Float	410236	
Analog Input Channel 09 Gain	403064	Float	410238	
Analog Input Channel 09 Offset	403066	Float	410240	
Analog Input Channel 10 Gain	403068	Float	410242	
Analog Input Channel 10 Offset	403070	Float	410244	
Analog Input Channel 11 Gain	403072	Float	410246	
Analog Input Channel 11 Offset	403074	Float	410248	
Analog Input Channel 12 Gain	403076	Float	410250	
Analog Input Channel 12 Offset	403078	Float	410252	
Analog Input Channel 13 Gain	403080	Float	410254	
Analog Input Channel 13 Offset	403082	Float	410256	
Analog Input Channel 14 Gain	403084	Float	410258	
Analog Input Channel 14 Offset	403086	Float	410260	
Analog Input Channel 15 Gain	403088	Float	410262	
Analog Input Channel 15 Offset	403090	Float	410264	
Analog Input Channel 16 Gain	403092	Float	410266	
Analog Input Channel 16 Offset	403094	Float	410268	

10.3 Identification block

Register addresses: 300001 – 300014
Concentrated ranges: 310033, 310135 – 310146 (mA)
311033, 311135 – 311146 (scaled)

10.3.1 Product Number

This register contains the product number of the PIO36 board and always contains the value 2210100 for the standard version.

10.3.2 Serial Number

This register contains the serial number of the PIO36 board.

10.3.3 Hardware Version

This register contains the revision number of the PCB-board.

10.3.4 Firmware Version

These registers contain the firmware version of the currently running firmware.

10.3.5 Firmware CRC

This register contains the CRC of the flash storage for the firmware.

10.3.6 Configuration CRC

This register contains the CRC of the values stored in the persistent storage.

10.3.7 Heartbeat

This register counts seconds since initialization is completed.

10.4 Configuration block

Register addresses: 300050 – 300099, 300117 – 300147 and 400050 – 400099, 400117 – 400147
Concentrated ranges: 310148 – 310211, 310214 – 310219, 310225 – 310231 (mA)
311148 – 311211, 311214 – 311219, 311225 – 311231 (scaled)
410044 – 410107, 410110 – 410114, 410120 – 410125 (write)

10.4.1 IPv4 address

This register specifies the IPv4 address the PIO36 uses on its ethernet interface.

This register is stored in the persistent storage and must be flushed if it is to be remembered at next power cycle or restart.

This register requires a power cycle or restart for changes to apply.

10.4.2 Subnet mask

This register specifies the subnet mask the PIO36 uses on its ethernet interface.

This register is stored in the persistent storage and must be flushed if it is to be remembered at next power cycle or restart.

This register requires a power cycle or restart for changes to apply.

10.4.3 Default gateway

This register specifies the default router the PIO36 uses on its ethernet interface.

This register is stored in the persistent storage and must be flushed if it is to be remembered at next power cycle or restart.

This register requires a power cycle or restart for changes to apply.

10.4.4 MAC address

These registers specify the MAC address for the PIO36 ethernet interface.

The first register contains the first two octets, the second the third and fourth octets and the last register the last two octets. For each register the MSB gives the first octet of its octets.

E.g., the MAC-address 0E-EC-A7-6D-9C-B5 is stored as the three Uint16 values 0x0EEC, 0xA76D and 0x9CB5.

This register is stored in the persistent storage and must be flushed if it is to be remembered at next power cycle or restart.

This register requires a power cycle or restart for changes to apply.

10.4.5 Baudrate RS232

This register specifies the baudrate given in bauds of the RS232 connection.

This register is stored in the persistent storage and must be flushed if it is to be remembered at next power cycle or restart.

This register requires a power cycle or restart for changes to apply.

10.4.6 Baudrate RS485

This register specifies the baudrate given in bauds of the RS485 connection.

This register is stored in the persistent storage and must be flushed if it is to be remembered at next power cycle or restart.

This register requires a power cycle or restart for changes to apply.

10.4.7 Modbus Base Address

This register specifies the base-address (before taking the switches into account) for the Modbus interface on the PIO36 board.

This register is stored in the persistent storage and must be flushed if it is to be remembered at next power cycle or restart.

This register requires a power cycle or restart for changes to apply.

10.4.8 Modbus Address

This register contains the actual address used for the Modbus interface.

This register is read-only and any attempt to read or write the associated holding register are ignored.

10.4.9 Lost-coms timeout

This register gives the lost-coms timeout value in seconds. A value of 0 disables the lost-coms alarm.

This register is stored in the persistent storage and must be flushed if it is to be remembered at next power cycle or restart.

10.4.10 Lost-coms Keep Value

This register specifies the lost-coms behaviour for the PWM outputs. Each bit in this register controls the behaviour of a different output, the least significant bit controls PWM 1 and the most significant bit control PWM 16.

If the associated bit is set, then the output keeps its current value in the case of a lost-coms alarm, otherwise the output value is set to the value specified in Lost-coms duty.

This register is stored in the persistent storage and must be flushed if it is to be remembered at next power cycle or restart.

10.4.11 Lost COM's output

These registers specify the lost COM's value for each output. A value of 0 (or less, including -inf and NaN) is fully off and a value of 100 (or more, including +inf) is fully on.

These registers are stored in the persistent storage and must be flushed if it is to be remembered at next power cycle or restart.

10.4.12 Power-on default output

These registers specify the power-on value for each output. A value of 0 (or less, including -inf and NaN) is fully off and a value of 100 (or more, including +inf) is fully on.

These registers are stored in the persistent storage and must be flushed if it is to be remembered at next power cycle or restart.

10.4.13 Sensor Power configuration

This register controls the lost-coms and power-on configuration for the Sensor power output. The register is a bitmask with 4 bits defined below.

This register is stored in the persistent storage and must be flushed if it is to be remembered at next power cycle or restart.

10.4.13.1 Bit 0: Ignore fault

If this bit is set, the sensor power output ignores the sensor power fault signal and allows to enable the output if the fault is set. As there are two separate transistors for the outputs and their faults are combined, this may be necessary if there is no load connected to one of them as the open-circuit detection will otherwise fault the output as long as it is off.

10.4.13.2 Bit 1: Lost-coms keep

If this bit is set, the sensor power output keeps its current value in the case of a lost-coms alarm, otherwise the Lost-coms Value is used.

10.4.13.3 Bit 2: Lost-coms Value

If Lost-coms keep is not set, this value is applied to the sensor power output in the case of a lost-coms alarm.

10.4.13.4 Bit 3: Power-on value

This value specifies the power-on value for the sensor power output.

10.4.14 Ethernet Port config

These registers control the speed of the two Ethernet ports. The register is a bitmask with 3 bits defined below. These registers are stored in the persistent storage and must be flushed if it is to be remembered at next power cycle or restart.

This register requires a power cycle or restart for changes to apply.

Value	Ethernet Port Configuration
0 (000)	Auto negotiate, 10 Mbps, full/half duplex
1 (001) [Default]	Auto negotiate, 10/100 Mbps, full/half duplex
2 (010)	Auto negotiate, 10 Mbps, half duplex
3 (011)	Auto negotiate, 10/100 Mbps, half duplex
4 (100)	Force, 10 Mbps, full duplex
5 (101)	Force, 100 Mbps, full duplex
6 (110)	Force, 10 Mbps, half duplex
7 (111)	Force, 100 Mbps, half duplex

10.4.14.1 Bit 0: Allow 100 Mbps

If this bit is set, the port will support 100 Mbps, otherwise only 10 Mbps can be used.

10.4.14.2 Bit 1: Only allow half-duplex

If this bit is set, the port will only allow half-duplex, otherwise both half- and full-duplex can be used.

10.4.14.3 Bit 2: Disable auto-negotiate (force speed/duplex)

If this bit is set auto-negotiate is disabled and the two previous bit set what speed/duplex to use.

10.4.15 RSTP enabled

Writing a non-zero value to this register will enable RSTP for the internal ethernet switch.

This register is stored in the persistent storage and must be flushed if it is to be remembered at next power cycle or restart.

10.5 Diagnostics block

Register addresses: 300100 – 300109 and 400102 – 400104
Concentrated ranges: 310034 – 310036, 310127 – 310133 (mA)
310034 – 310036, 310127 – 310133 (scaled)
410033 – 410035 (write)

10.5.1 Uptime

This register gives the time since power-on in seconds.

10.5.2 System Warnings

This register is a bitmask of warnings from the system. Only the input register contains the actual values if read, and to clear the warnings write the corresponding bit in the Reset System Warnings holding register to 1. Writing a value of 0 to a bit has no effect.

10.5.2.1 Bit 0: Unexpected restart

This bit is set if the onboard watchdog has restarted the program.

10.5.2.2 Bit 1: Hardware fault

This bit is set if the CPU fails initialize one or more of its internal or external peripherals.

10.5.2.3 Bit 2: Persistent storage fault

This bit is set if one or more persistent values failed to load from the persistent storage.

10.5.3 Sensor Alarms

This register is a bitmask of alarms from the sensors. Only the input register contains the actual values if read, and to clear the alarms write the corresponding bit in the Reset Sensor Alarms holding register to 1. Writing a value of 0 to a bit has no effect.

10.5.3.1 Bit 0: Water Alarm

This bit is set if the water probe has detected water.

10.5.3.2 Bit 1: Low supply voltage

This bit is set if the PIO36 supply voltage falls below its warning threshold.

10.5.3.3 Bit 2: High supply voltage

This bit is set if the PIO36 supply voltage exceeds its warning threshold.

10.5.3.4 Bit 3: High supply current

This bit is set if the supply current exceeds its warning threshold.

10.5.3.5 Bit 4: High CPU temperature

This bit is set if the temperature of the CPU gets too high.

10.5.3.6 Bit 5: Low Pt100 temperature

This bit is set if the measured temperature of the Pt100 temperature sensor falls below the given threshold.

10.5.3.7 Bit 6: High Pt100 temperature

This bit is set if the measured temperature of the Pt100 temperature sensor goes above the given threshold.

10.5.3.8 Bit 7: Sensor power fault

This bit is set if the sensor power driver detects a fault situation, either a short circuit or an open circuit.

10.5.3.9 Bit 8: Lost-coms

This bit is set if the PIO36 is without Modbus communication for at least the number of seconds given in the Lost-coms timeout register.

10.5.4 PWM Faults

This register is a bitmask of PWM output faults. Only the input register contains the actual values to read, and to clear the faults write the corresponding bit in the Reset PWM Faults holding register to 1. Writing a value of 0 to a bit has no effect.

Bit 0 – 15 corresponds to PWM output fault 1 – 16.

10.5.5 Modbus Packet Count

This register contains the number of valid Modbus packets received that are addressed to this unit.

10.5.6 Modbus Dropped Packet Count

This register contains the number of Modbus packets dropped due to bad CRC or other malformed packets. It does not count the number of valid packets received that are not addressed to this unit.

10.5.7 EEPROM write counter

This register contains the number of write cycles to the persistent storage.

10.6 System commands block

Register addresses: 300111 – 300116 and 400111 – 400116
Concentrated ranges: 310032 (mA)
311032 (scaled)
410032, 410036 – 410037 (write)

10.6.1 Sensor Power Enable

This register controls and show the state of the Sensor Power output. Writing a non-zero value to this register will enable the sensor power output (unless it is faulted and the ignore fault flag is not set).

10.6.2 Restart command

Writing a non-zero value to this register will restart the pio36 board.

10.6.3 EEPROM flush command

Writing a non-zero value to this register will flush any pending changes down to the persistent storage.

10.6.4 EEPROM cleanup command

Writing a non-zero value to this register will prune any unused values from the persistent storage. This is normally not needed but may be necessary after a firmware update.

10.6.5 RS232 terminal mode

Writing a non-zero value to this register will set the RS232 interface into terminal mode. If this interface is used for Modbus communication, it will respond to its current packet before switching.

10.6.6 RS485 terminal mode

Writing a non-zero value to this register will set the RS485 interface into terminal mode. If this interface is used for Modbus communication, it will respond to its current packet before switching.

10.7 PWM block

Register addresses: 301000 – 301136 and 401000 – 401136
Concentrated ranges: 310000 – 310030, 310213 (mA)
311000 – 311030, 311213 (scaled)
410000 – 410030 (write)

10.7.1 PWM duty

These registers control the duty of the PWM outputs. A value of 0 (or less, including -inf and NaN) is fully off and a value of 100 (or more, including +inf) is fully on.

10.7.2 PWM synchronization command

Writing a non-zero value to this register synchronizes the dither counters.

10.7.3 PWM Ignore faults

This register is a bitmask for the 16 PWM output, bit 0 – 15 corresponds to PWM output 1 – 16. If the corresponding bit is set that output ignores the fault bit, preventing the output to be disabled if a fault occurs and allows enabling it if a fault has not been cleared.

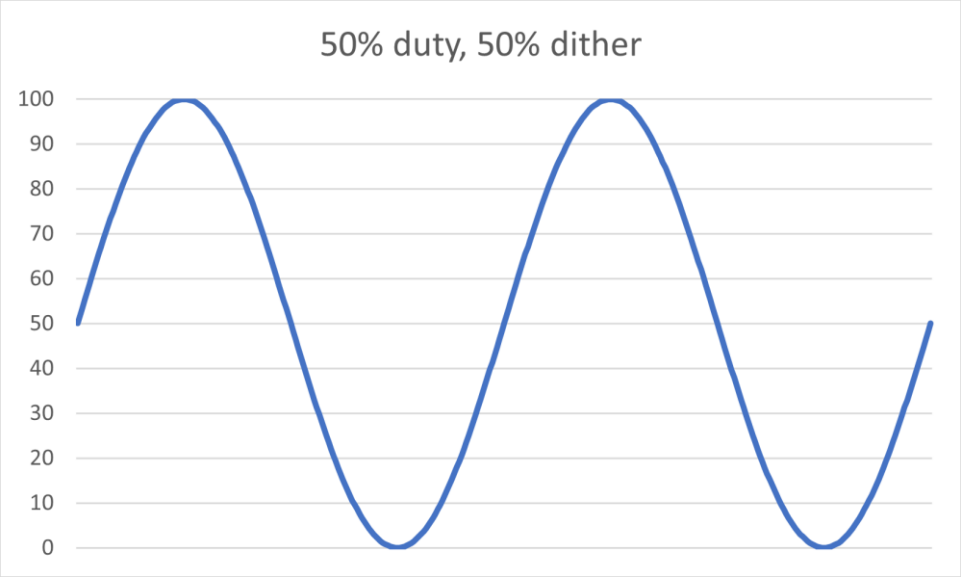
10.7.4 PWM Frequency

These registers control the PWM frequency of the outputs in Hz. The outputs are grouped by 4 which must have the same PWM frequency. Values written to these registers are limited (and clipped to if outside) to the range 1 – 500 Hz.

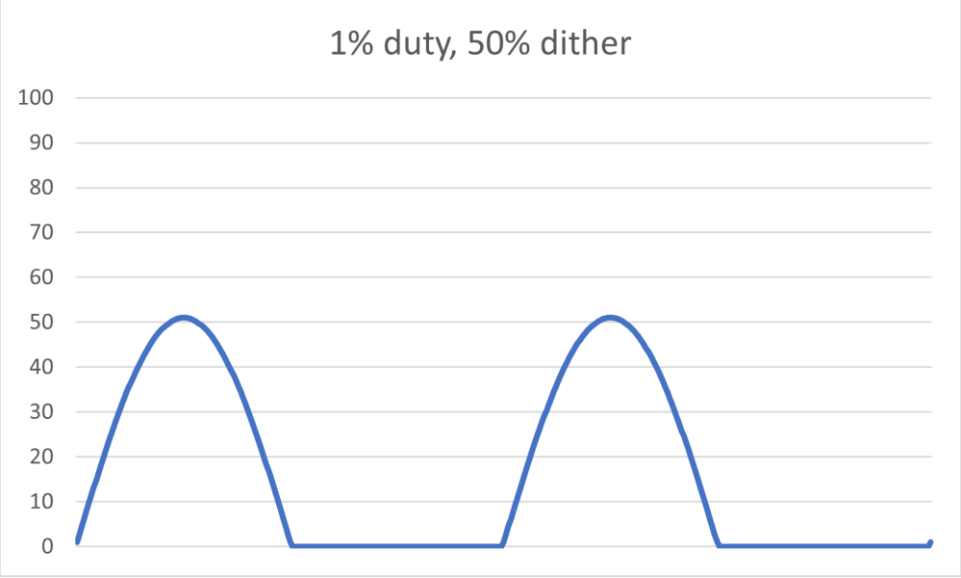
10.7.5 PWM dither duty

These registers control the dither of each output. The value is given the same as for PWM duty but represent the amplitude of a sine wave that modifies the PWM duty. The combined duty is clipped the same way as the PWM duty is.

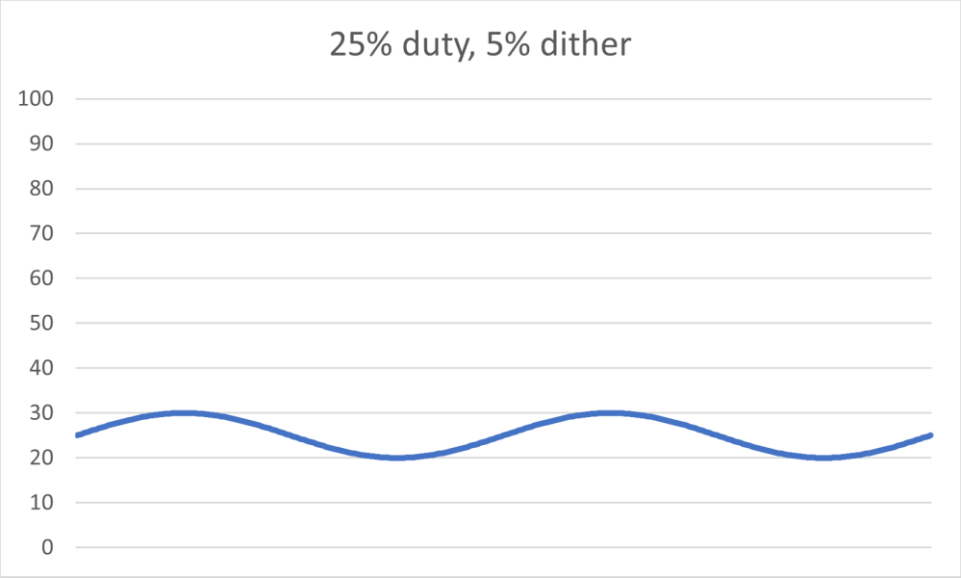
Note: If the PWM duty is set to 0 or below or to 100 or above, the dither is disabled.



Example 1: A PWM duty of 50 and a PWM dither duty of 50 results roughly in a sine wave with peak to peak of 0 – 100.



Example 2: A PWM duty of 1 and a PWM dither duty of 50.

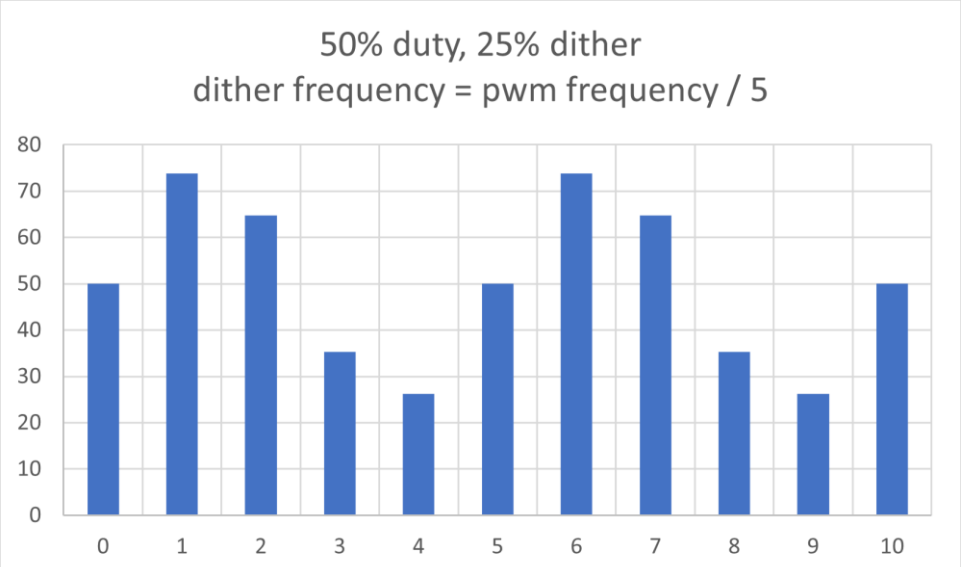


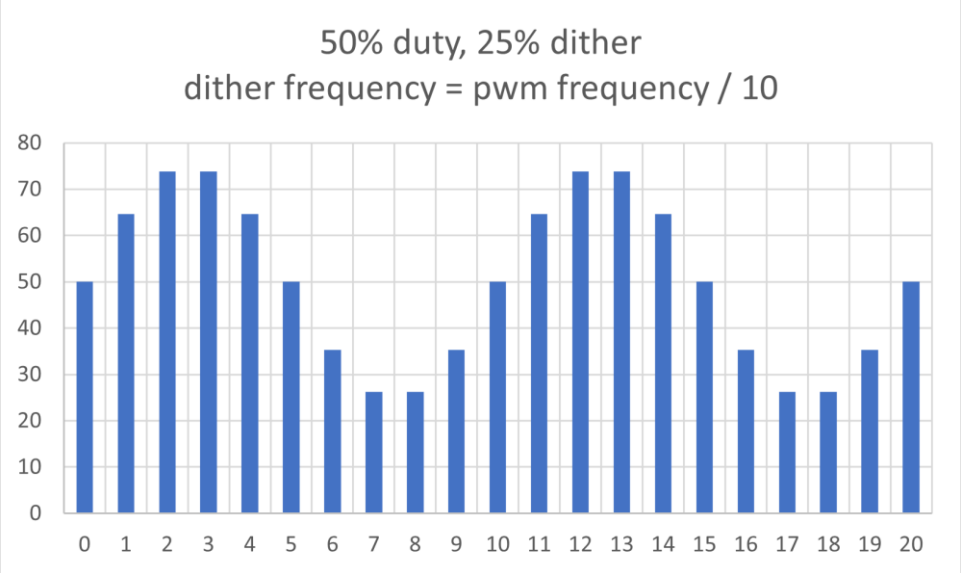
Example 3: A PWM duty of 25 and a PWM dither duty of 5.

10.7.6 PWM Dither Frequency

These registers control the PWM dither frequency in Hz. The dither is calculated every on every PWM update, thus dither frequencies above the Nyquist of the PWM frequency will not work. The ratio between the PWM frequency and the dither frequency gives the number of updates per dither period.

Example: With the dither frequency at a tenth of the PWM frequency the sine wave shape is more pronounced than when the dither frequency is a fifth of the PWM frequency.





10.7.7 PWM dither phase

These registers control the PWM dither phase offsets in degrees (0 – 360).

10.8 Sensors block

Register addresses: 302000 – 302120 and 402082 – 402100
Concentrated ranges: 310037 – 310125 (mA)
311037 – 311125 (scaled)
410038 – 410043 (write)

10.8.1 Supply voltage

This register contains the measured supply voltage in volt.

10.8.2 Supply current

This register contains the measured current in Ampere to the PWM outputs and the sensor power output.

10.8.3 CPU temperature

This register contains the measured CPU temperature in degrees Celsius.

10.8.4 Pt100 sensor

This register contains the measured temperature from the Pt100 sensor in degrees Celsius.

10.8.5 Analog Input Channels (mA/V)

These registers contain the measured current in mA. The first four channels can also measure voltage, dependent on switches on the board. If the switch is set in voltage mode, the current register will read as NaN otherwise if the switch is set to measure current the voltage register will read as NaN.

10.8.6 Analog Input voltage flags

This register is a bitmask with bit 0 to 3 showing if Analog Input Channel 1 to 4 is in voltage measuring mode. If the bit is set the channel measures voltage, otherwise it measures current.

10.8.7 Analog Input Channel (Scaled)

These registers contain the scaled value of the Analog input channels.

10.8.8 Digital Input levels

This register contains the current logical level of the digital inputs. Bit 0 to 3 corresponds to digital input 1 to 4.

10.8.9 Digital Input Counts

These registers contain the number of rising edges detected on the digital input. Reset the count back to 0 by writing a non-zero value to the holding register.

These registers may only be read through the input register and will always read as zero from the holding registers.

10.8.10 Digital Input Frequency

These registers contain the calculated frequency in Hz of the digital inputs.

10.8.11 Encoder 1 count

This register contains the gray2 count of combining digital input channel 1 and 2. Writing a non-zero value to the holding register resets the count back to 0.

This register may only be read through the input register and will always read as zero from the holding register.

10.8.12 Encoder 2 count

This register contains the gray2 count of combining digital input channel 3 and 4. Writing a non-zero value to the holding register resets the count back to 0.

This register may only be read through the input register and will always read as zero from the holding register.

10.8.13 Encoder Frequency

These registers contain the frequency in Hz of the encoder inputs.

10.8.14 Flow

These registers contain the flow values calculated from the digital input frequency from a flowmeter.

10.8.15 Accumulated Flow

These registers contain the accumulated flow values calculated from the digital input counts. These registers are zeroed when the corresponding digital input count is reset.

10.9 Sensor configuration block

Register addresses: 303000 – 303144 and 403000 – 403144
Concentrated ranges: 310232 – 310374 (mA)
311232 – 311374 (scaled)
410126 – 410268 (write)

10.9.1 Analog input range, gain and offset

These registers define the scaling values for the 4 to 20 mA sensor on the associated analogue input channel.

The values are scaled using the following formula:

$$Ain = \frac{mA}{20 - 4} * range * gain + offset$$

These registers are stored in the persistent storage and must be flushed if it is to be remembered at next power cycle or restart.

10.9.2 Pt100 gain and offset

These registers define calibration values for the Pt100 sensor.

These registers are stored in the persistent storage and must be flushed if it is to be remembered at next power cycle or restart.

10.9.3 Flow K-factors

These registers give the K-factors in pulses per litre for scaling the flow measurements.

These registers are stored in the persistent storage and must be flushed if it is to be remembered at next power cycle or restart.

10.9.4 Analog Input Filter Constants

These registers define the filter for the analogue inputs. The filter is a simple IIR filter and the value is the fraction of previous measurement to use. A value of 0 results in an unfiltered value being used, and a value of 1 means the value will not be updated. Values below 0 or at and above 1 should not be used.

These registers are stored in the persistent storage and must be flushed if it is to be remembered at next power cycle or restart.

10.9.5 Pt100 Filter constant

This register defines the filter for the Pt100 sensor. The filter is a simple IIR filter and the value is the fraction of previous measurement to use. A value of 0 results in an unfiltered value being used, and a value of 1 means the value will not be updated. Values below 0 or at and above 1 should not be used.

These registers are stored in the persistent storage and must be flushed if it is to be remembered at next power cycle or restart.

10.9.6 Pt100 Low Limit and Pt100 High Limit

These registers define the alarm thresholds for respectively low and high Pt100 temperature alarms.

These registers are stored in the persistent storage and must be flushed if it is to be remembered at next power cycle or restart.

10.10 Calibration block

Register addresses: 304000 – 304129, 304133
Concentrated ranges: -

10.10.1 Analog input raw value

These registers contain the raw value from the analogue input ADC.

10.10.2 Analog Input mA Gain and Offset

These registers contain the calibration values for the milliamperes measurement on the analogue inputs.

10.10.3 Analog Input V Gain and Offset

These registers contain the calibration values for the voltage measurements on the analogue inputs.

10.10.4 Supply Voltage Gain and Offset

These registers contain the calibration values for the supply voltage measurement.

10.10.5 Supply Current Gain and Offset

These registers contain the calibration values for the supply current measurement.

10.10.6 CPU Temperature Gain and Offset

These registers contain the calibration values for the CPU temperature measurement.

10.10.7 Raw Water Alarm value

This register contains the raw ADC value for the water intrusion detector. This register should not be used for detecting an alarm, use the Water Alarm bit in Sensor Alarms for that.

10.10.8 WA pin time

This register contains the time in milliseconds the water intrusion detector probes for water.

10.10.9 WA wait time

This register contains the time in milliseconds between activating the water detector.

10.10.10 WA Limits

These registers define the allowed range for the water probe, used for triggering the WA alarm. The default settings are 0 for low-limit and 800 for high-limit.

10.10.11 Supply Voltage Limits

These registers define the allowed range for the supply voltage, used for triggering the supply voltage alarms.

10.10.12 Supply Current Limits

These registers define the allowed range for the supply current, used for triggering the supply current alarm.

10.10.13 CPU Temperature Limits

These registers define the allowed range for the CPU Temperature, used for triggering the CPU temperature alarm.

10.10.14 Supply Voltage Filter Constant

This register defines the filter for the supply voltage. The filter is a simple IIR filter, and the value is the fraction of previous measurement to use.

10.10.15 Supply Current Filter Constant

This register defines the filter for the supply current. The filter is a simple IIR filter, and the value is the fraction of previous measurement to use.

10.10.16 CPU Temperature Filter Constant

This register defines the filter for the CPU temperature. The filter is a simple IIR filter, and the value is the fraction of previous measurement to use.

Appendix A – Declaration of Conformity

SAMSVARSERKLÆRING
DECLARATION OF CONFORMITY

INNOVA AS erklærer på eget ansvar, at følgende produkt:
INNOVA AS declares under its sole responsibility that the following product:

PRODUKTNAVN / PRODUCT NAME	*PIO36*
MODELL / TYPE	2210100

Som dekkes av denne erklæringen, er i overensstemmelse med:
Is covered by this declaration, and is in accordance to following requirements:

Electromagnetic Compatibility (EMC) Directive 2014/30/EU

Som implementert i:
As implemented through:

IEC60533 – Electrical and electronic installations in ships – Electromagnetic compatibility
IEC610000-6-4 – Electromagnetic compatibility (EMC): Generic Standards – Emission standard for industrial environments
IEC610000-6-2 – Electromagnetic compatibility (EMC): Generic Standards – Immunity for industrial environments

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ANSVARLIG SIGNATUR/
RESPONSIBLE SIGNATURE:

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